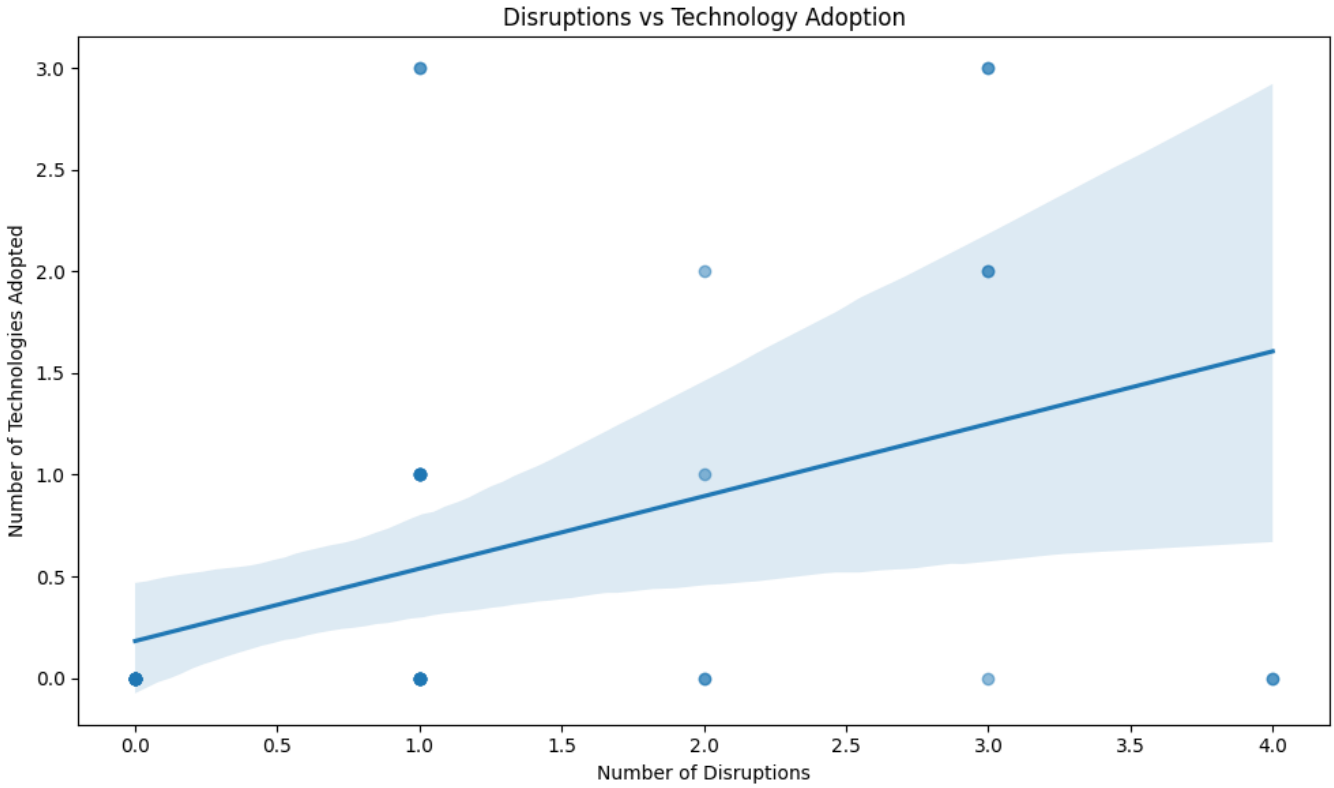
Hypothesis H1a

Companies that have experienced higher levels of supply chain disruptions are more likely to adopt resilience-enhancing technologies.

Correlation Analysis

Pearson Correlation: r = 0.418, p = 0.004

Significant positive correlation found between disruption count and technology adoption.



Group Comparison

Mann-Whitney U Test: $U = 253.0$, $p = 0.038$

Significant difference in technology adoption between high and low disruption companies.

Descriptive Statistics

	High Disruption	Low Disruption
count	11.000000	34.000000
mean	1.181818	0.382353
std	1.250454	0.779071
min	0.000000	0.000000
25%	0.000000	0.000000
50%	1.000000	0.000000
75%	2.000000	0.750000
max	3.000000	3.000000

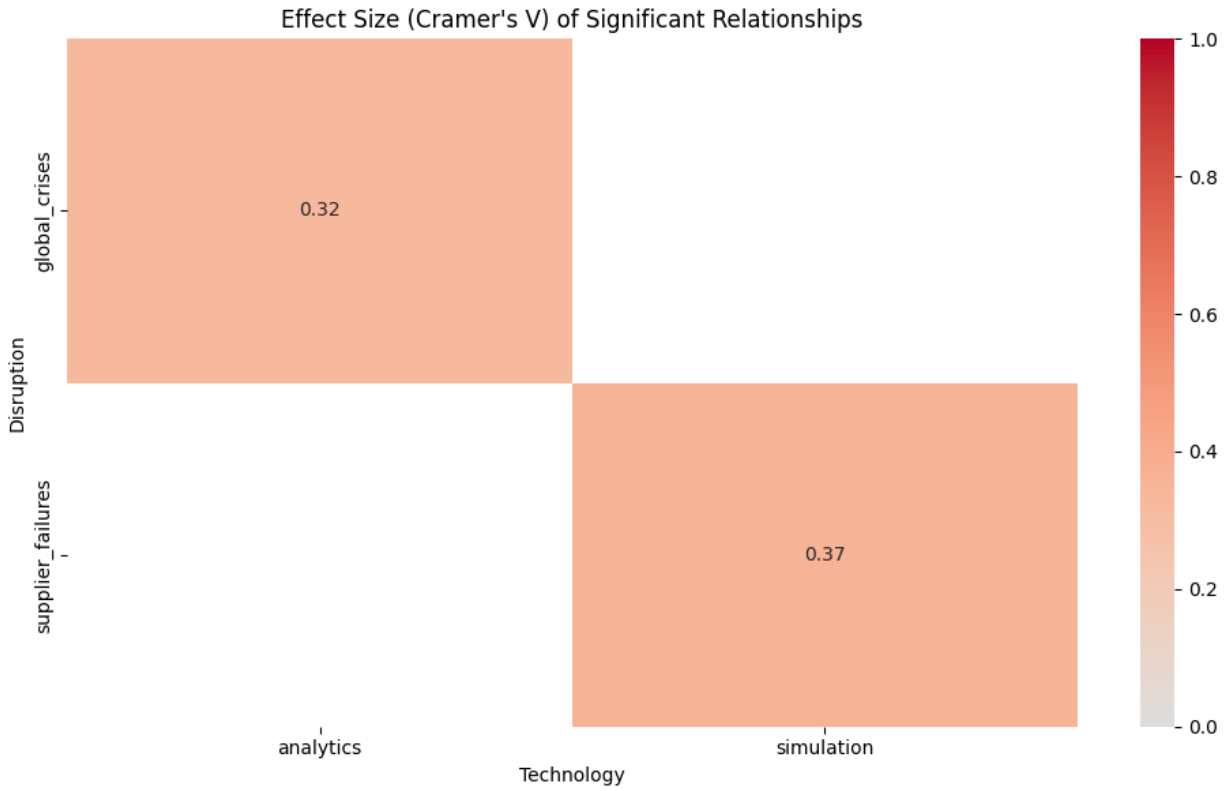
Hypothesis H1b

The impact of disruptions significantly influences technology choice.

Disruption-Technology Relationships

	Technology	Disruption	Chi-square	p-value	Cramer's V
0	ai	natural_disasters	0.075585	0.783372	0.040984
1	ai	supplier_failures	1.824467	0.176782	0.201355
2	ai	cybersecurity_threats	0.000000	1.000000	0.000000
3	ai	global_crises	2.501570	0.113733	0.235776
4	ai	transportation_delays	2.564464	0.109290	0.238722
5	iot	natural_disasters	0.083661	0.772396	0.043118
6	iot	supplier_failures	1.168244	0.279762	0.161124
7	iot	cybersecurity_threats	0.070689	0.790336	0.039634
8	iot	global_crises	2.326488	0.127189	0.227376
9	iot	transportation_delays	0.000000	1.000000	0.000000
10	blockchain	natural_disasters	0.000000	1.000000	0.000000
11	blockchain	supplier_failures	0.000000	1.000000	0.000000
12	blockchain	cybersecurity_threats	0.000000	1.000000	0.000000
13	blockchain	global_crises	0.000000	1.000000	0.000000
14	blockchain	transportation_delays	0.000000	1.000000	0.000000
15	computing	natural_disasters	0.000000	1.000000	0.000000
16	computing	supplier_failures	0.000000	1.000000	0.000000
17	computing	cybersecurity_threats	0.000000	1.000000	0.000000
18	computing	global_crises	0.000000	1.000000	0.000000
19	computing	transportation_delays	0.000000	1.000000	0.000000
20	analytics	natural_disasters	0.009502	0.922348	0.014531
21	analytics	supplier_failures	0.769637	0.380329	0.130779
22	analytics	cybersecurity_threats	0.840320	0.359305	0.136652
23	analytics	global_crises	4.674531	0.030613	0.322302

24	analytics	transportation_delays	0.856438	0.354737	0.137956
25	simulation	natural_disasters	0.083661	0.772396	0.043118
26	simulation	supplier_failures	6.007153	0.014248	0.365366
27	simulation	cybersecurity_threats	0.070689	0.790336	0.039634
28	simulation	global_crises	2.326488	0.127189	0.227376
29	simulation	transportation_delays	1.609464	0.204567	0.189119



Hypothesis H2a

The adoption rate of AI, IoT, and Big Data Analytics is higher in large enterprises compared to small and medium enterprises (SMEs).

Adoption Rates by Company Size

company_size Large

SME

which_technologies_does_your_company_currently_use_for_supply_chain_resilience_select_all_that_apply_artificial_intelligence_ai 0.166667 0.375

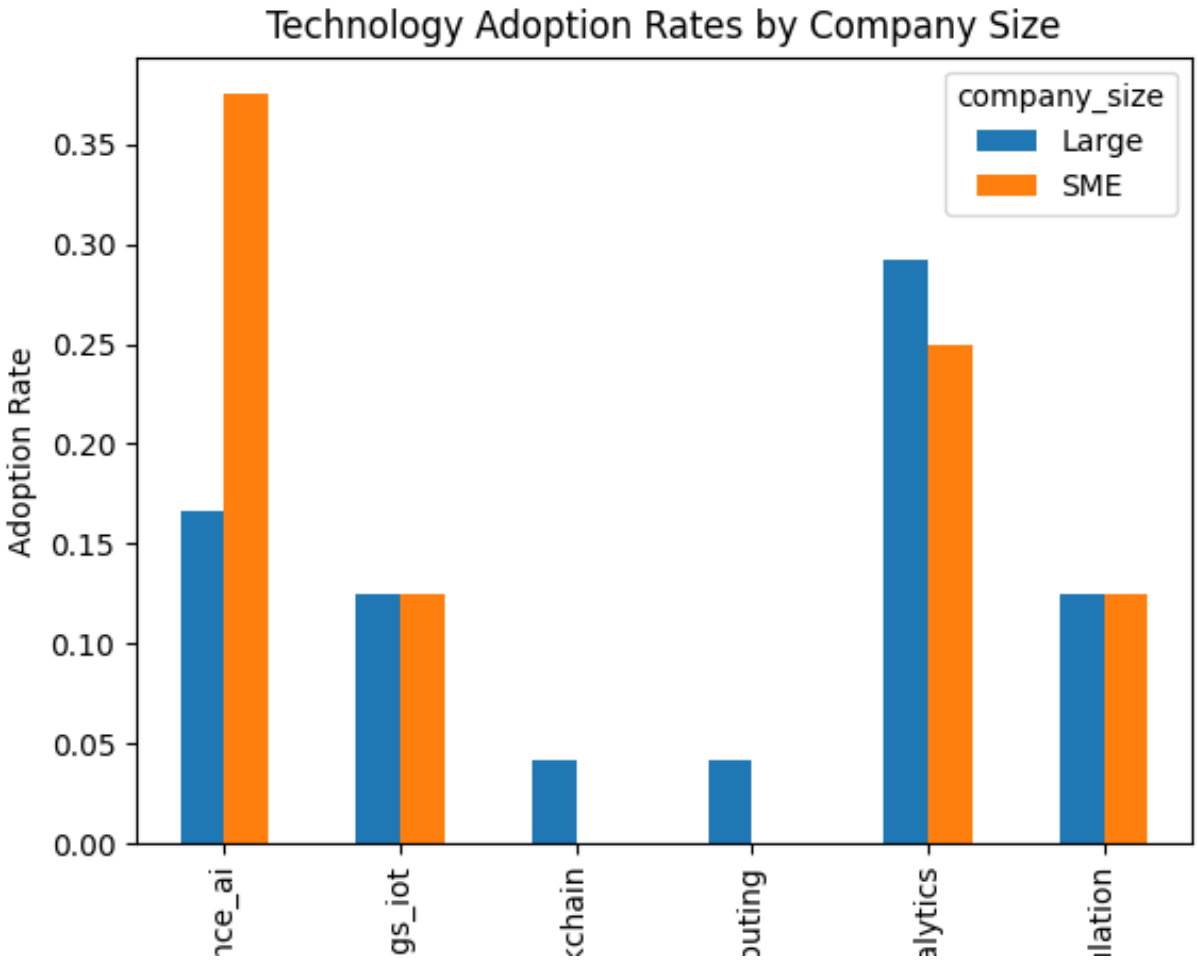
which_technologies_does_your_company_currently_use_for_supply_chain_resilience_select_all_that_apply_internet_of_things_iot 0.125000 0.125

which_technologies_does_your_company_currently_use_for_supply_chain_resilience_select_all_that_apply_blockchain 0.041667 0.000

which_technologies_does_your_company_currently_use_for_supply_chain_resilience_select_all_that_apply_cloud_computing 0.041667 0.000

which_technologies_does_your_company_currently_use_for_supply_chain_resilience_select_all_that_apply_big_data_analytics 0.291667 0.250

which_technologies_does_your_company_currently_use_for_supply_chain_resilience_select_all_that_apply_simulation 0.125000 0.125


Technology Adoption Rates by Company Size
company_size
Large
SME
Adoption Rate
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
nce_ai
gs_iot
cchain
puting
alytics
lation

Statistical Test for ai

ai:

Fisher's Exact Test: OR = 3.00, p = 0.327

No significant difference in adoption rates.

Statistical Test for iot

iot:

Fisher's Exact Test: OR = 1.00, p = 1.000

No significant difference in adoption rates.

Statistical Test for blockchain

blockchain:

Fisher's Exact Test: OR = 0.00, p = 1.000

No significant difference in adoption rates.

Statistical Test for computing

computing:

Fisher's Exact Test: OR = 0.00, p = 1.000

No significant difference in adoption rates.

Statistical Test for analytics

analytics:

Fisher's Exact Test: OR = 0.81, p = 1.000

No significant difference in adoption rates.

Statistical Test for simulation

simulation:

Fisher's Exact Test: OR = 1.00, p = 1.000

No significant difference in adoption rates.

Hypothesis H2b

Companies in highly disrupted industries (e.g., automotive, electronics) are more likely to implement Cloud Computing and Blockchain for resilience.

Adoption Rates by Industry

industry_group Manufacturing

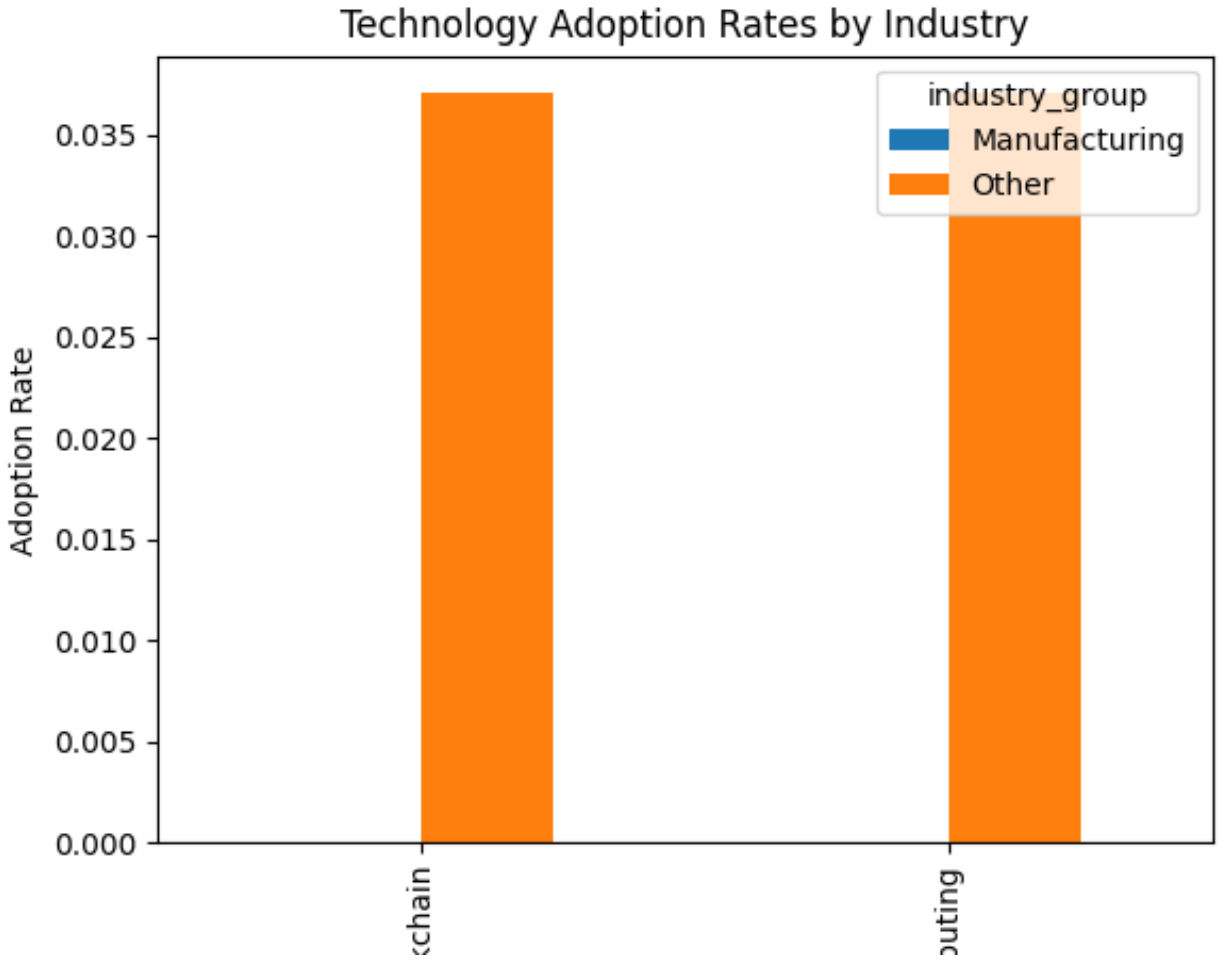
Other

which_technologies_does_your_company_currently_use_for_supply_chain_resilience_select_all_th

at_apply_blockchain 0.0 0.037037

which_technologies_does_your_company_currently_use_for_supply_chain_resilience_select_all_th

at_apply_cloud_computing 0.0 0.037037



Statistical Test for blockchain

blockchain:

Fisher's Exact Test: OR = inf, p = 1.000

No significant difference in adoption rates.

Statistical Test for computing

computing:

Fisher's Exact Test: OR = inf, p = 1.000

No significant difference in adoption rates.

Hypothesis H3a

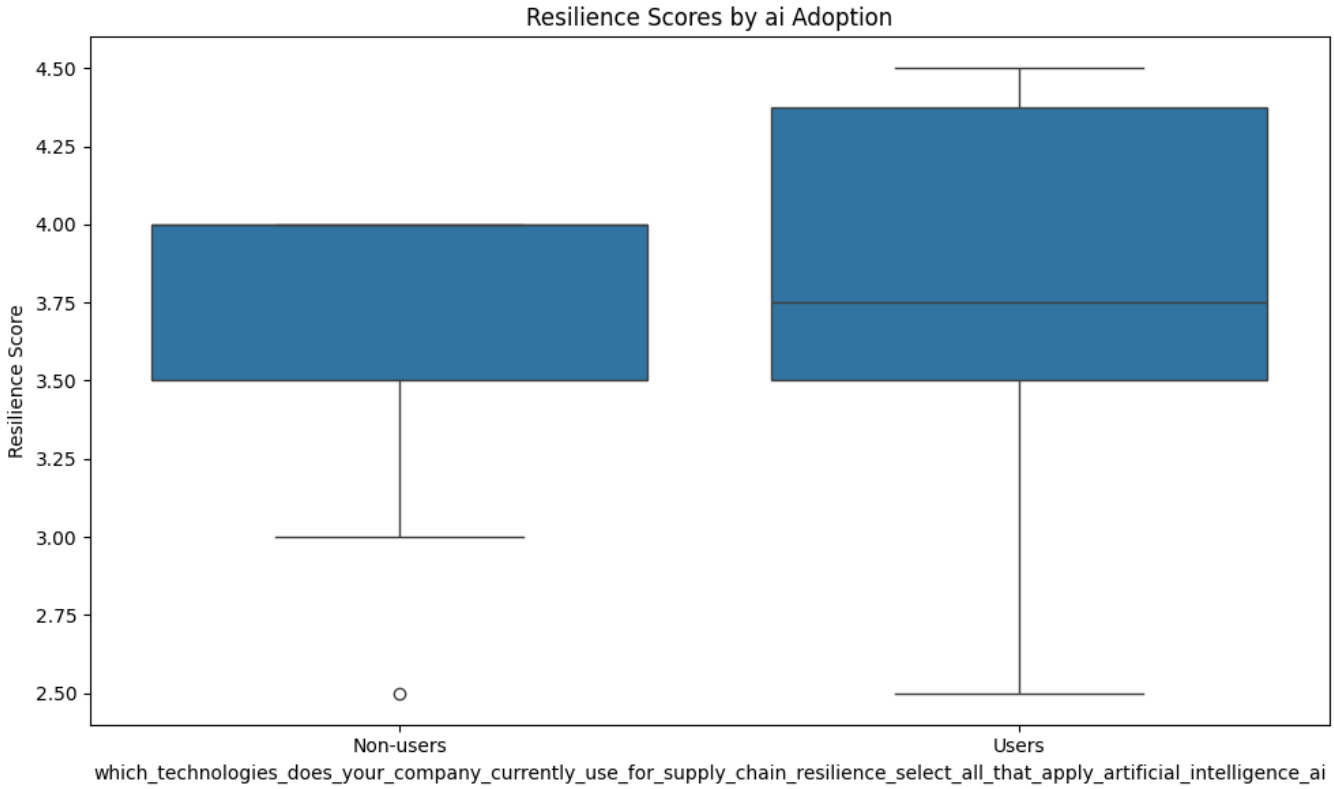
Companies using AI and IoT report higher supply chain efficiency and resilience compared to those not using them.

Effectiveness of ai

ai:

Mann-Whitney U = 55.0, p = 0.790

No significant difference in resilience scores.

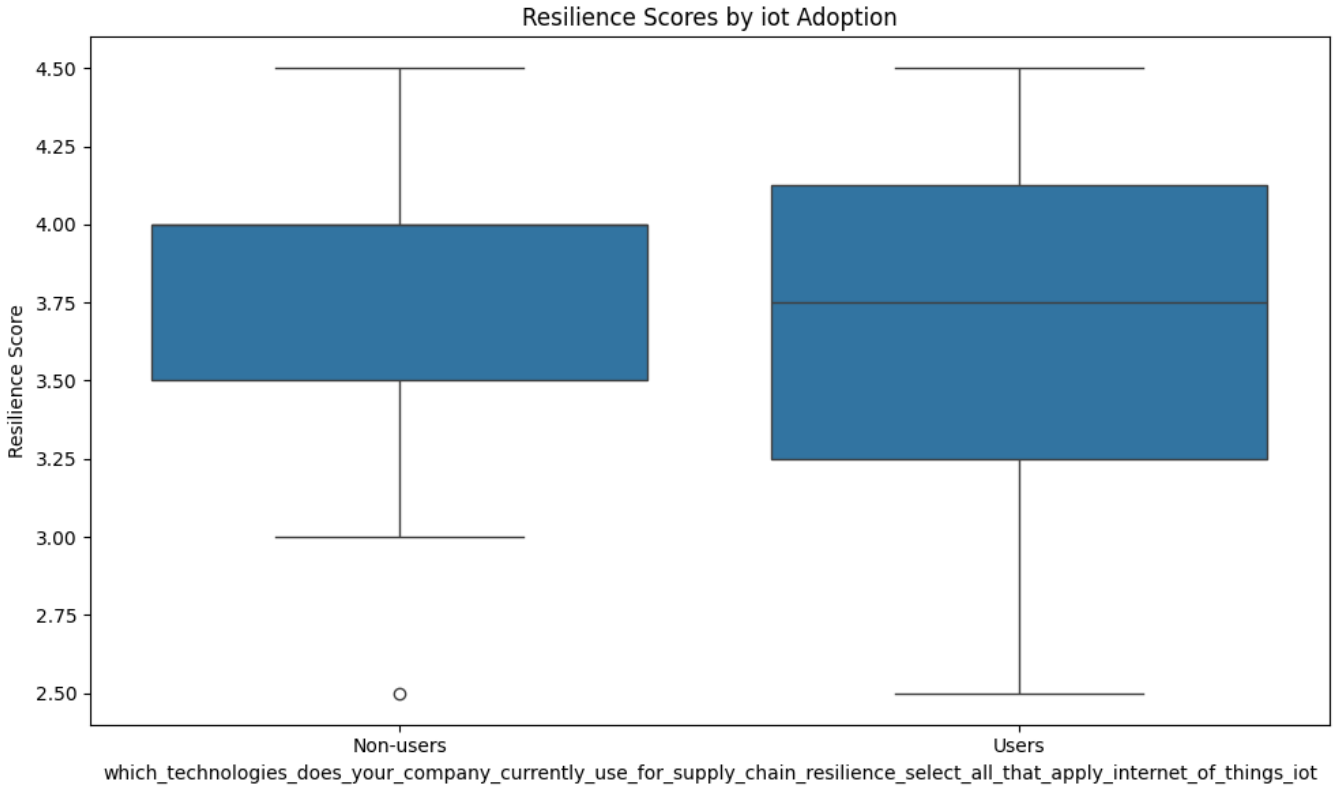


Effectiveness of iot

iot:

Mann-Whitney U = 36.5, p = 0.930

No significant difference in resilience scores.

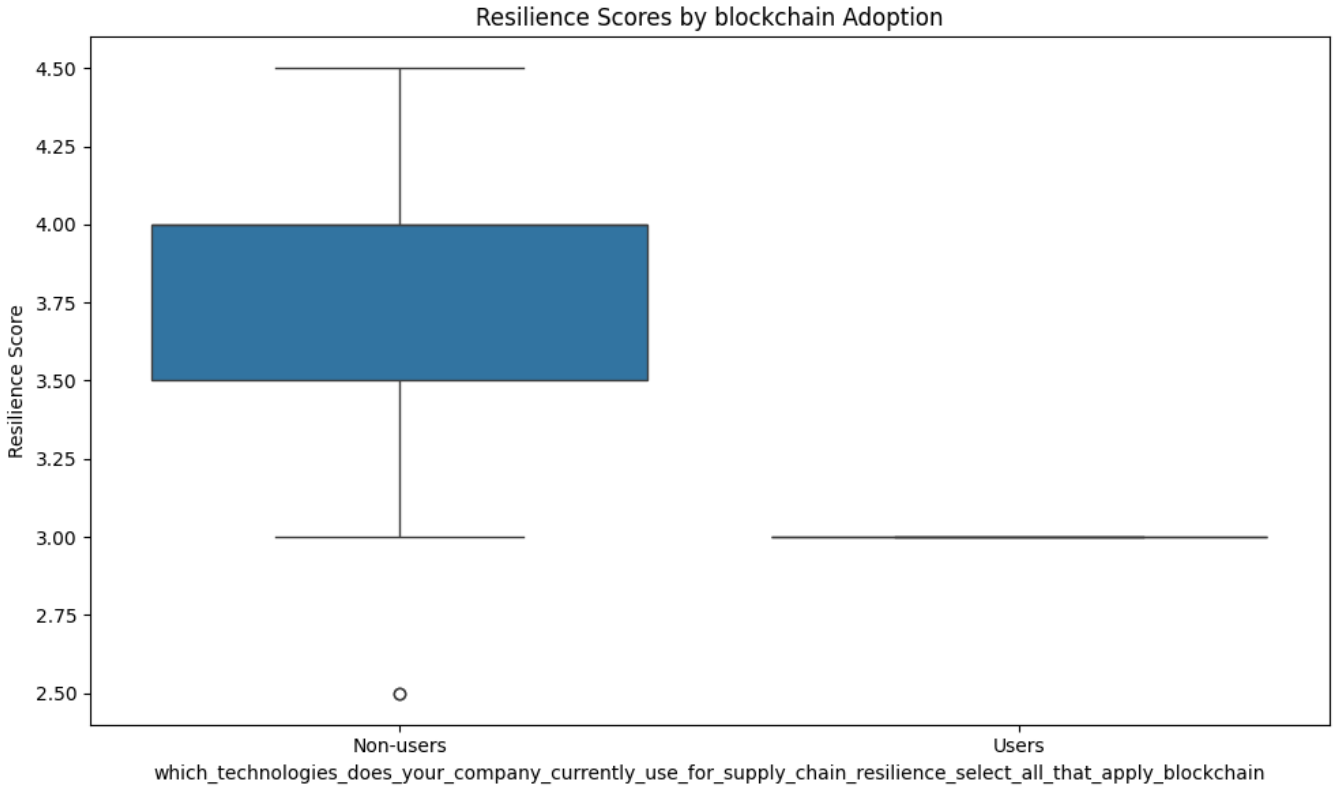


Effectiveness of blockchain

blockchain:

Mann-Whitney U = 2.5, p = 0.190

No significant difference in resilience scores.

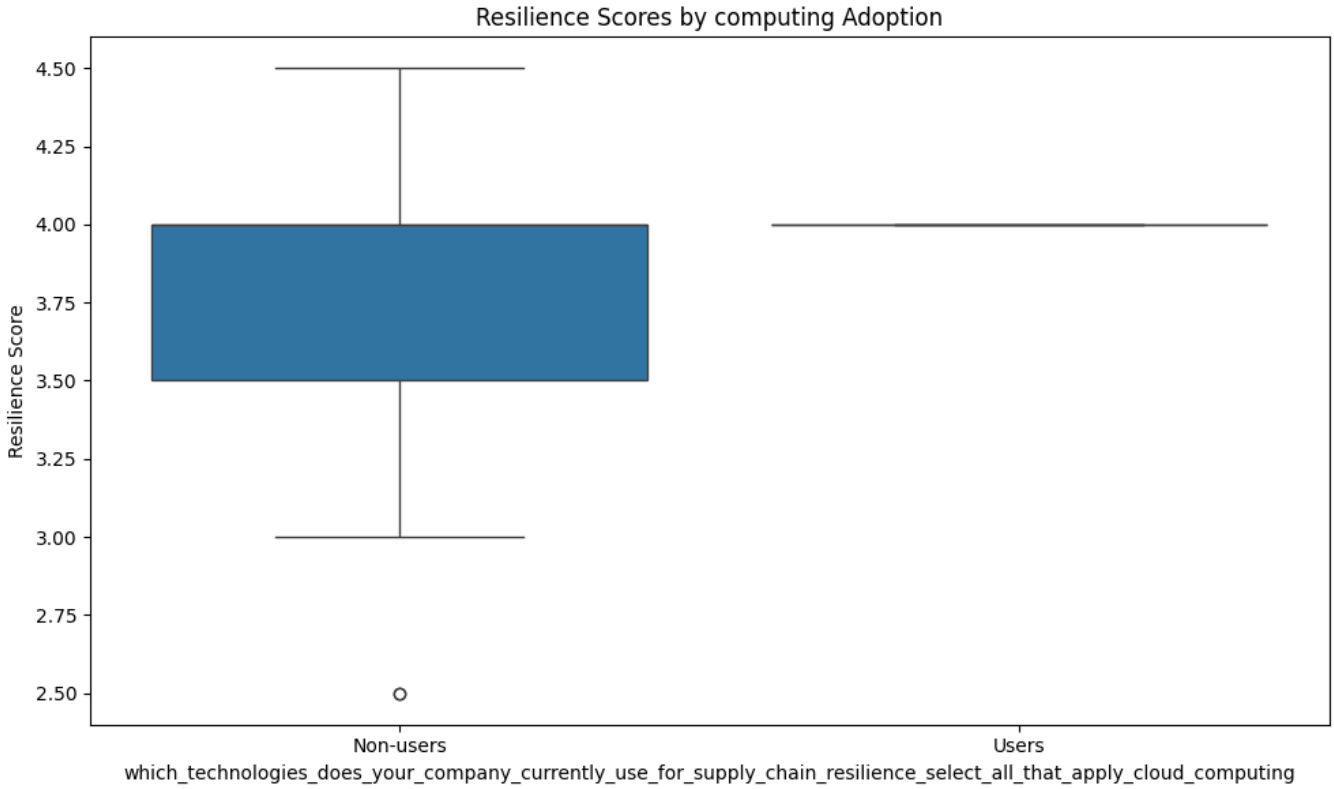


Effectiveness of computing

computing:

Mann-Whitney U = 14.5, p = 0.623

No significant difference in resilience scores.

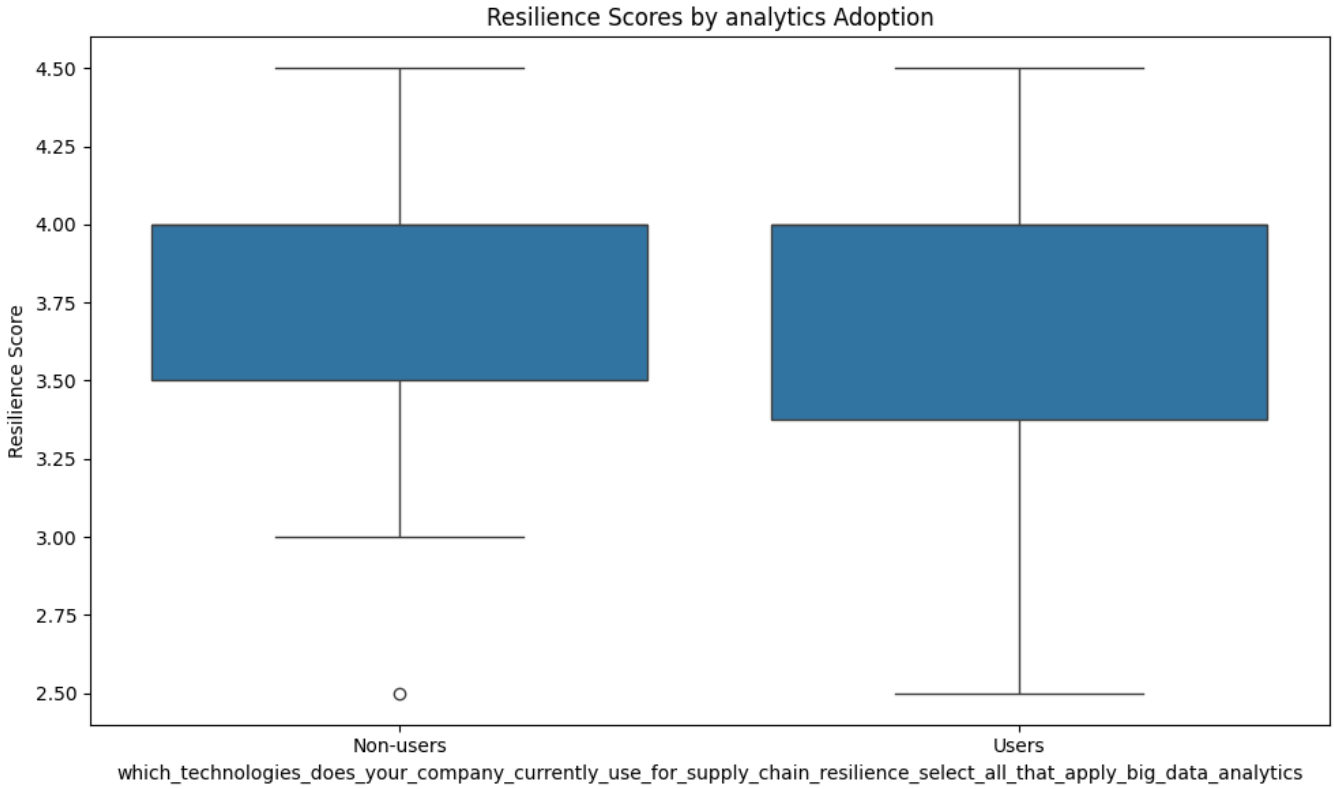


Effectiveness of analytics

analytics:

Mann-Whitney U = 60.5, p = 1.000

No significant difference in resilience scores.

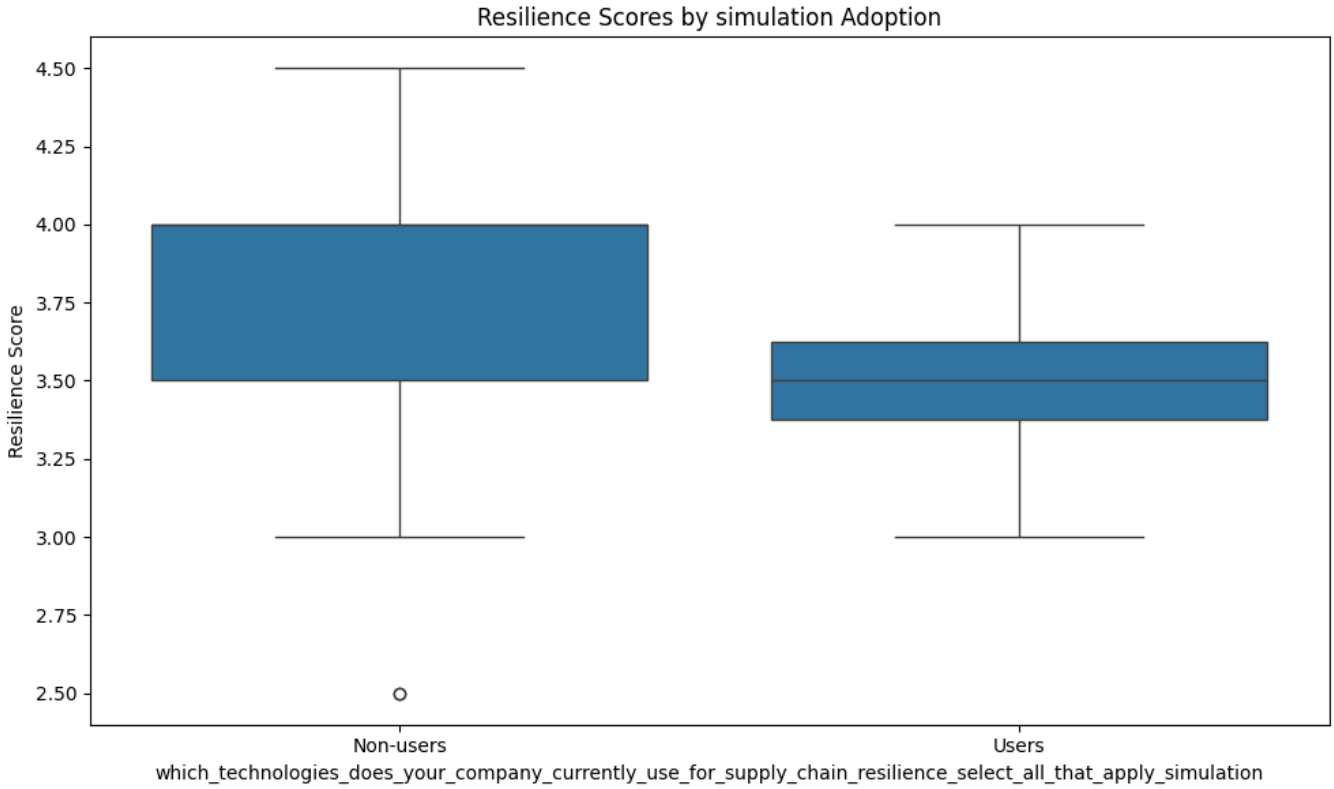


Effectiveness of simulation

simulation:

Mann-Whitney U = 23.0, p = 0.201

No significant difference in resilience scores.



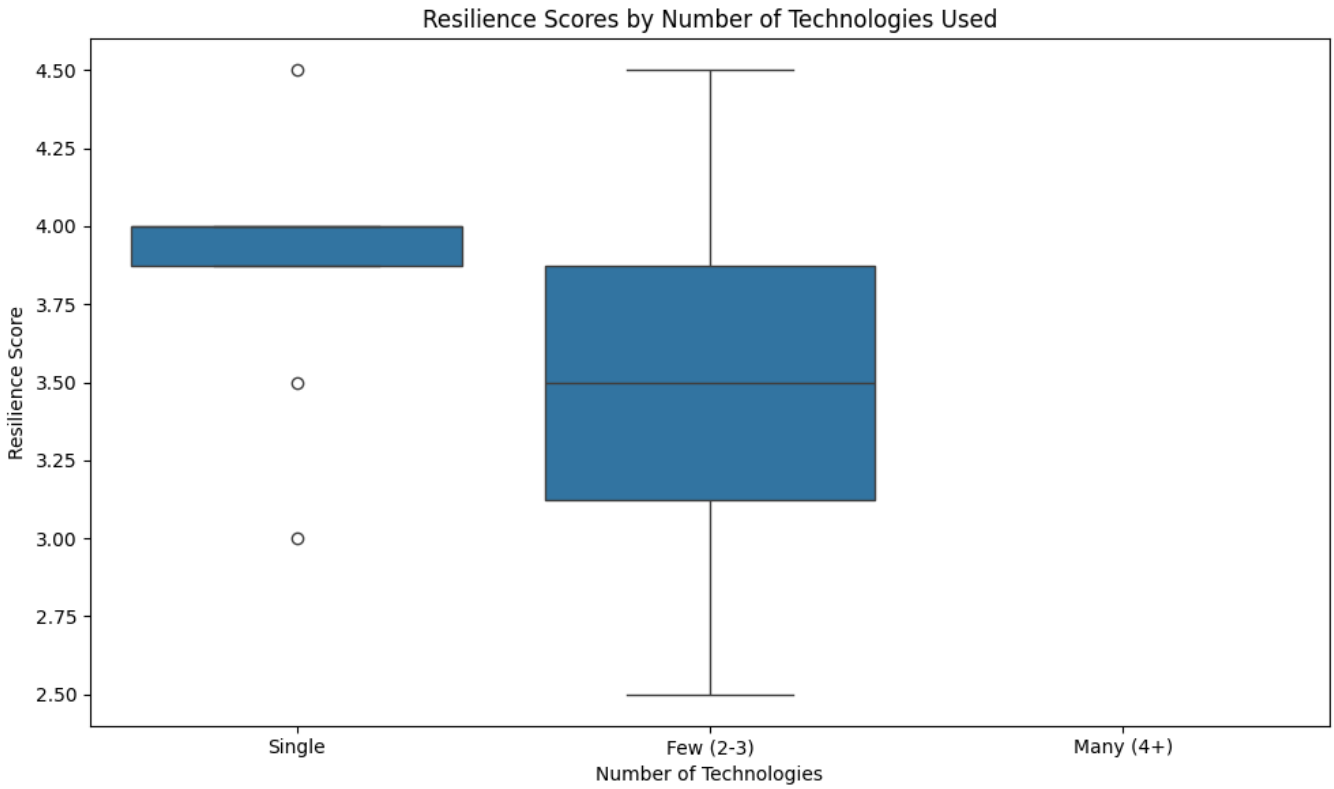
Hypothesis H3b

Companies using multiple technologies in combination report better resilience scores than those using a single technology.

Statistical Test Results

Kruskal-Wallis Test: H = nan, p = nan

No significant differences between technology groups.



Hypothesis H4a

High implementation costs are the primary barrier to adopting new supply chain technologies.

Barrier Analysis

Proportion reporting high implementation costs as barrier: 0.22

All Barriers:

what_are_the_biggest_barriers_preventing_further_adoption_of_these_technologies_select_all_that_apply_technical_challenges 0.266667

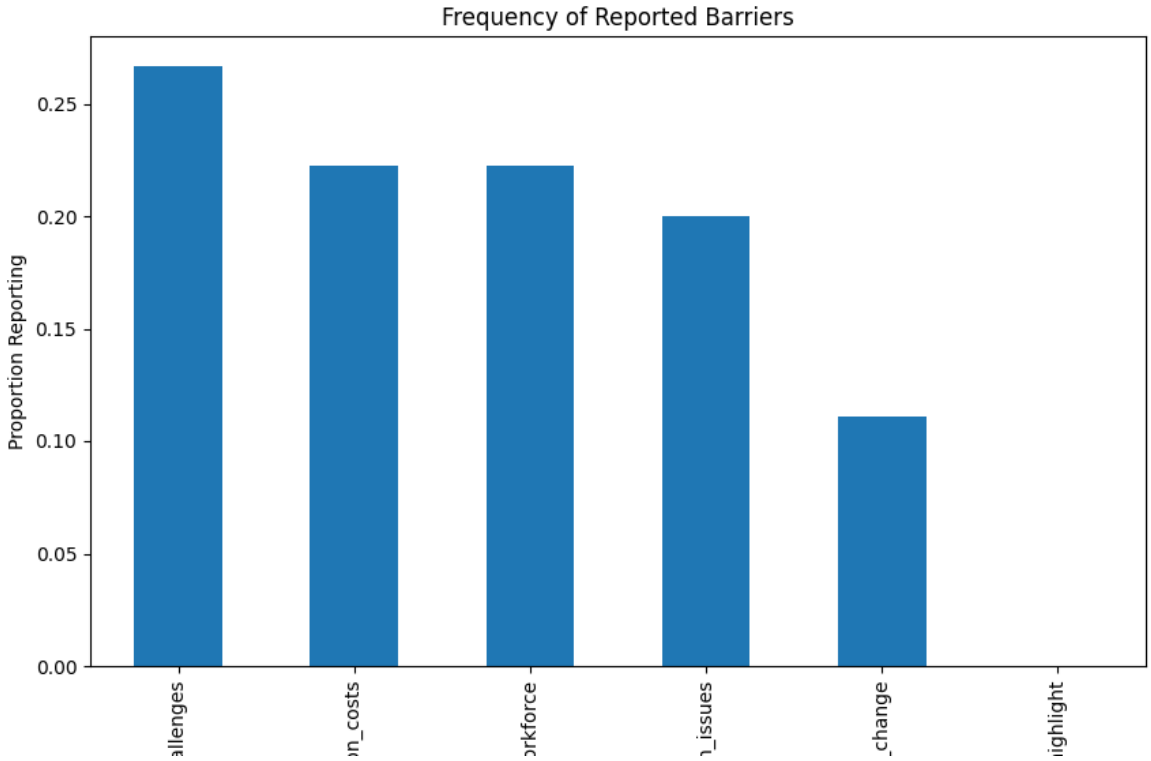
what_were_the_main_challenges_in_implementing_these_technologies_select_all_that_apply_high_implementation_costs 0.222222

what_were_the_main_challenges_in_implementing_these_technologies_select_all_that_apply_lack_of_skilled_workforce 0.222222

what_were_the_main_challenges_in_implementing_these_technologies_select_all_that_apply_integration_issues 0.200000

what_were_the_main_challenges_in_implementing_these_technologies_select_all_that_apply_resistance_to_change 0.111111

are_there_any_additional_decision_making_challenges_or_priorities_you_would_like_to_highlight NaN


Frequency of Reported Barriers
Proportion Reporting
0.25
0.20
0.15
0.10
0.05
0.00
allenges
on_costs
orkforce
n_issues
_change
ighlight

Hypothesis H4b

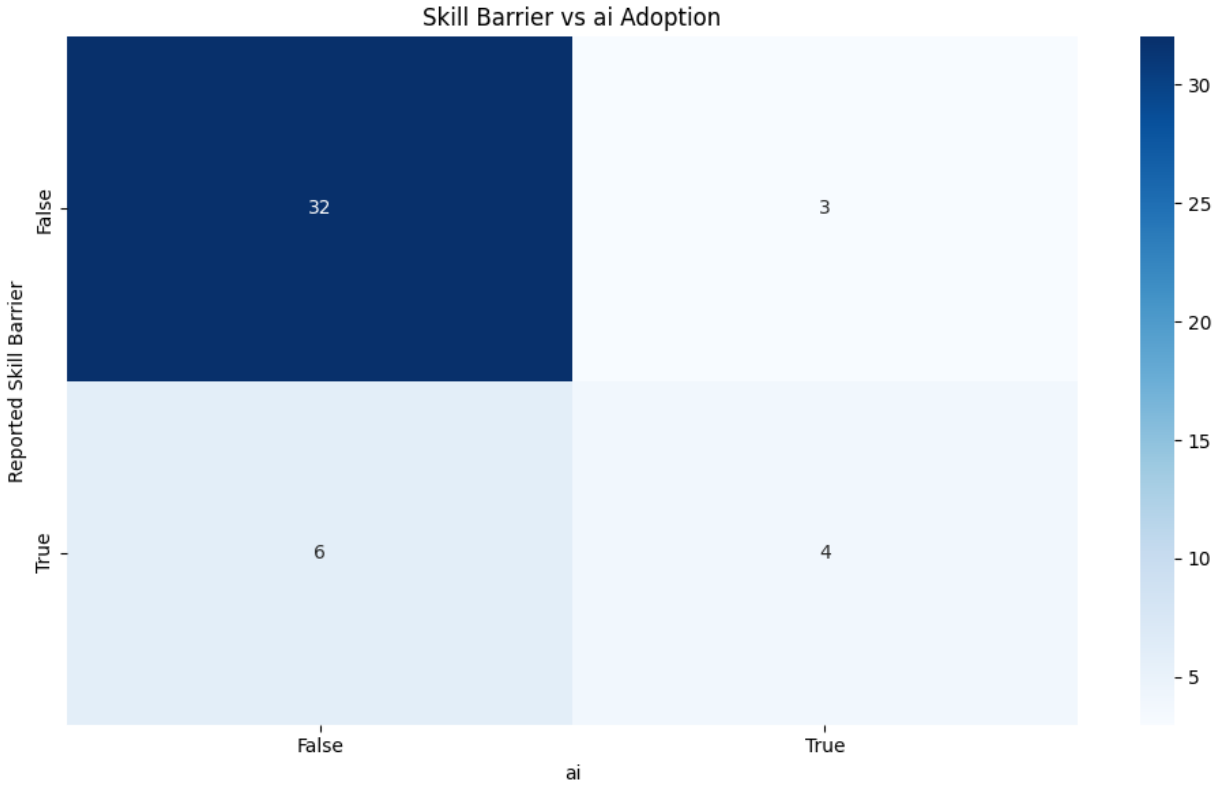
Lack of skilled workforce significantly affects the adoption of AI and Big Data Analytics in supply chain management.

Skill Barrier and ai

ai:

Fisher's Exact Test: OR = 7.11, p = 0.034

Significant relationship between skill barrier and adoption.

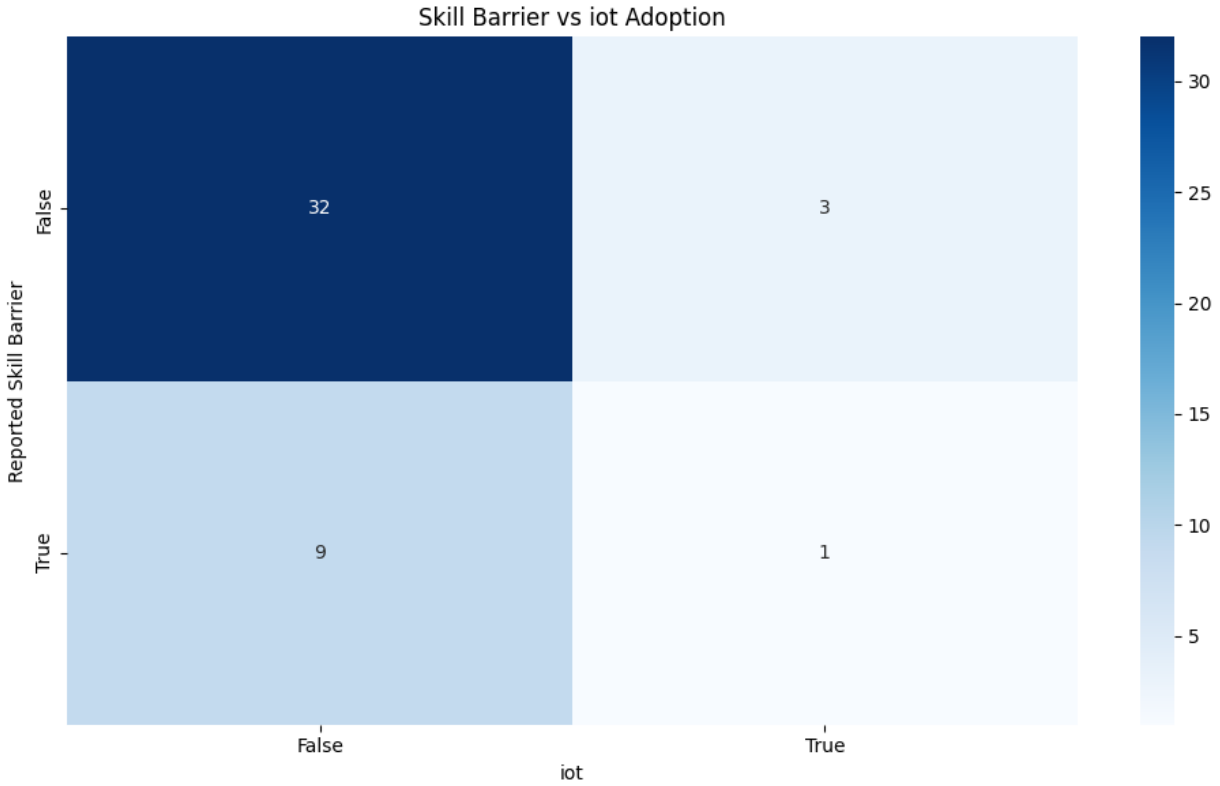


Skill Barrier and iot

iot:

Fisher's Exact Test: OR = 1.19, p = 1.000

No significant relationship found.

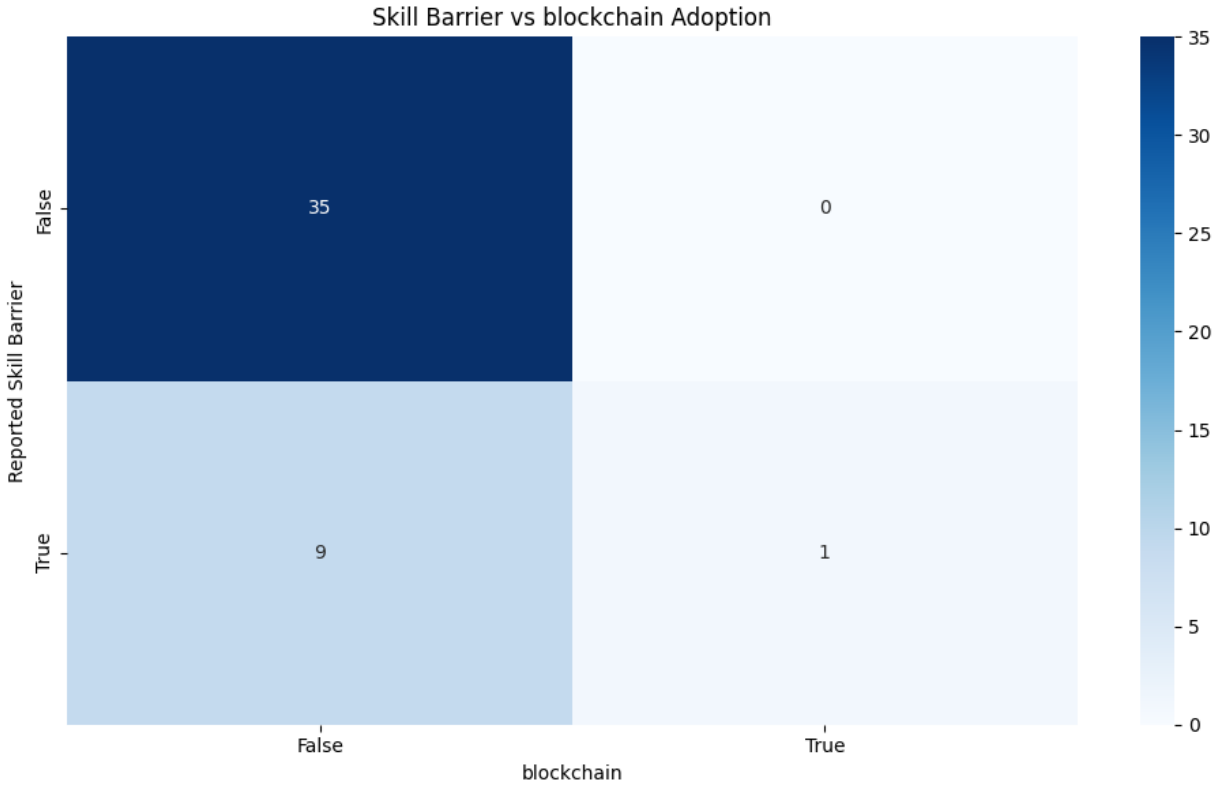


Skill Barrier and blockchain

blockchain:

Fisher's Exact Test: OR = inf, p = 0.222

No significant relationship found.

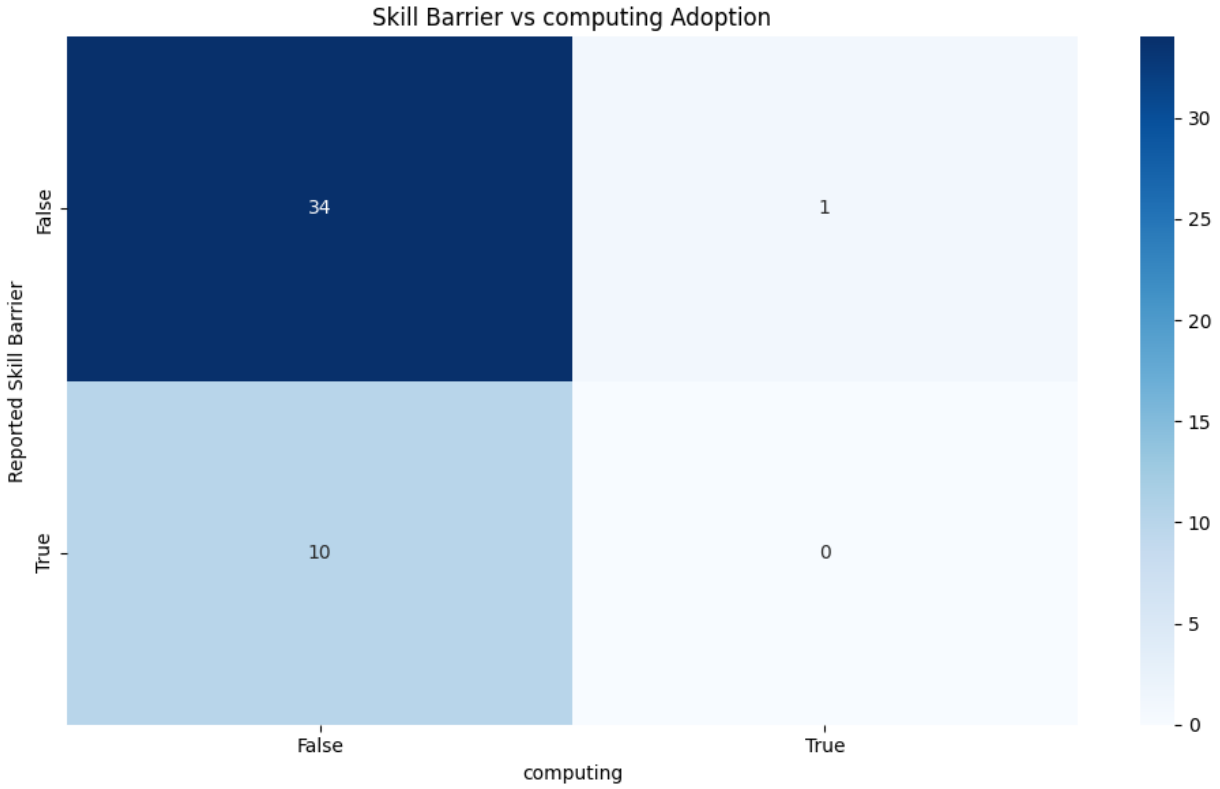


Skill Barrier and computing

computing:

Fisher's Exact Test: OR = 0.00, p = 1.000

No significant relationship found.

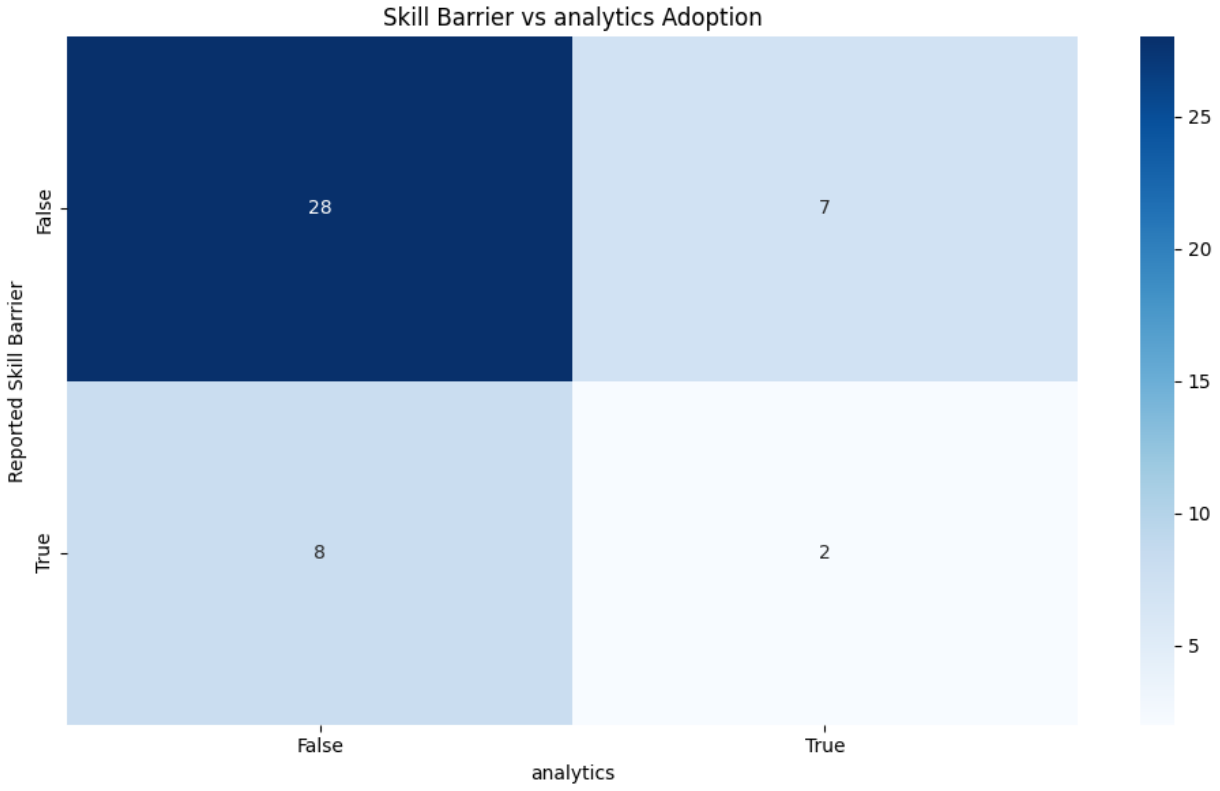


Skill Barrier and analytics

analytics:

Fisher's Exact Test: OR = 1.00, p = 1.000

No significant relationship found.

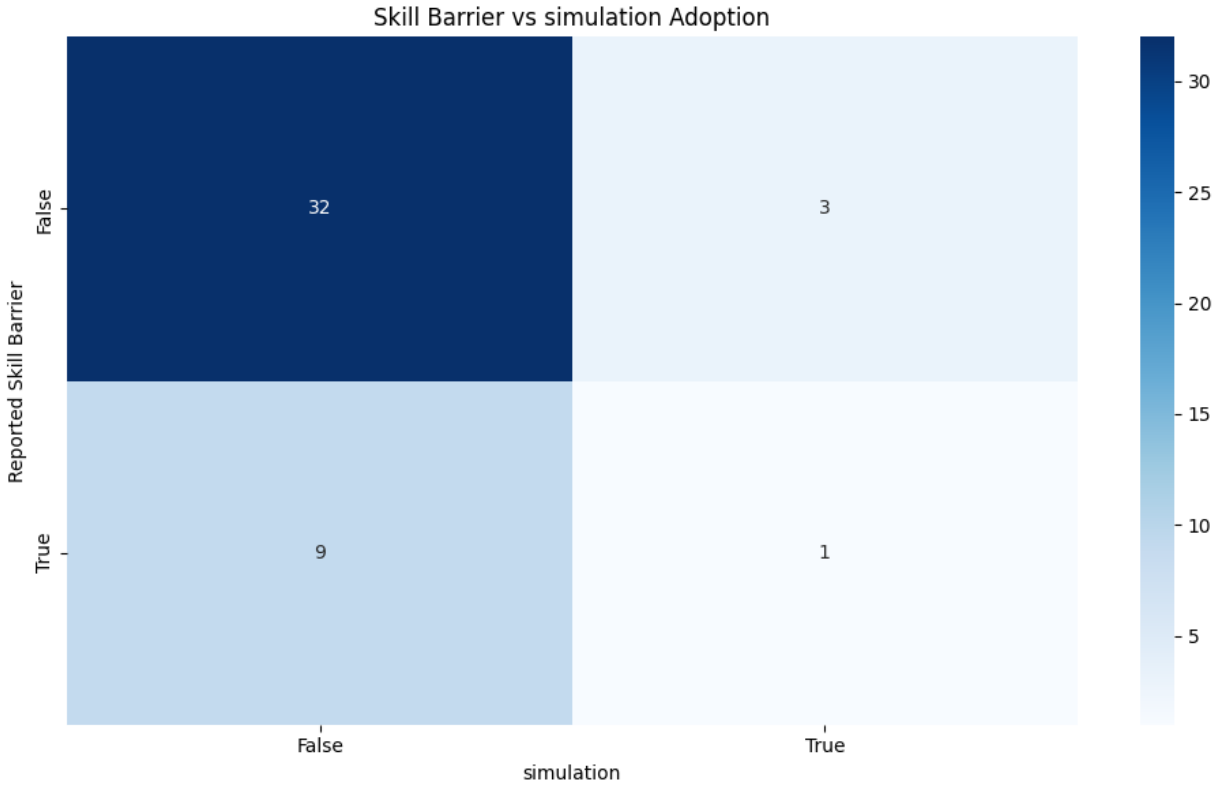


Skill Barrier and simulation

simulation:

Fisher's Exact Test: OR = 1.19, p = 1.000

No significant relationship found.



Hypothesis H5a

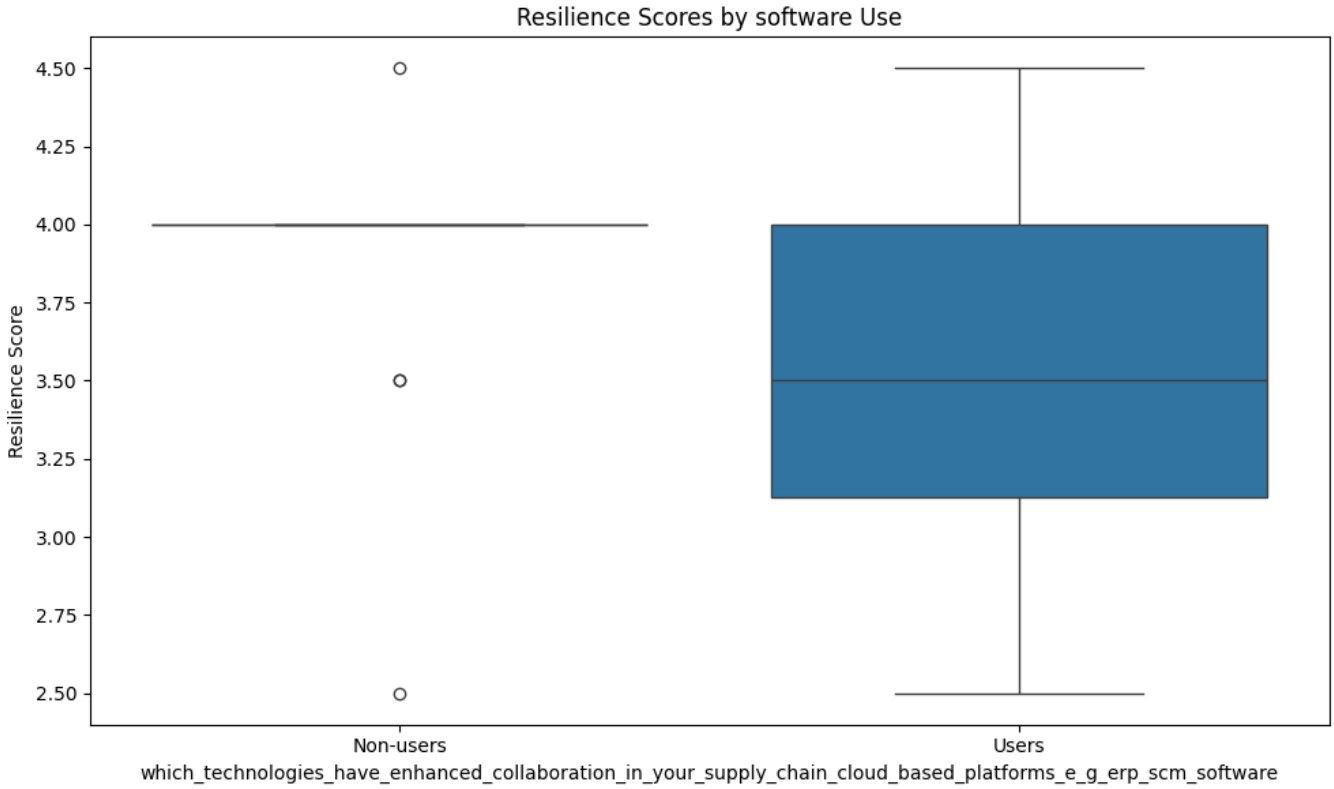
Companies that actively collaborate using digital platforms experience higher supply chain resilience.

Collaboration Technology: software

software:

Mann-Whitney U = 43.5, p = 0.157

No significant difference in resilience scores.

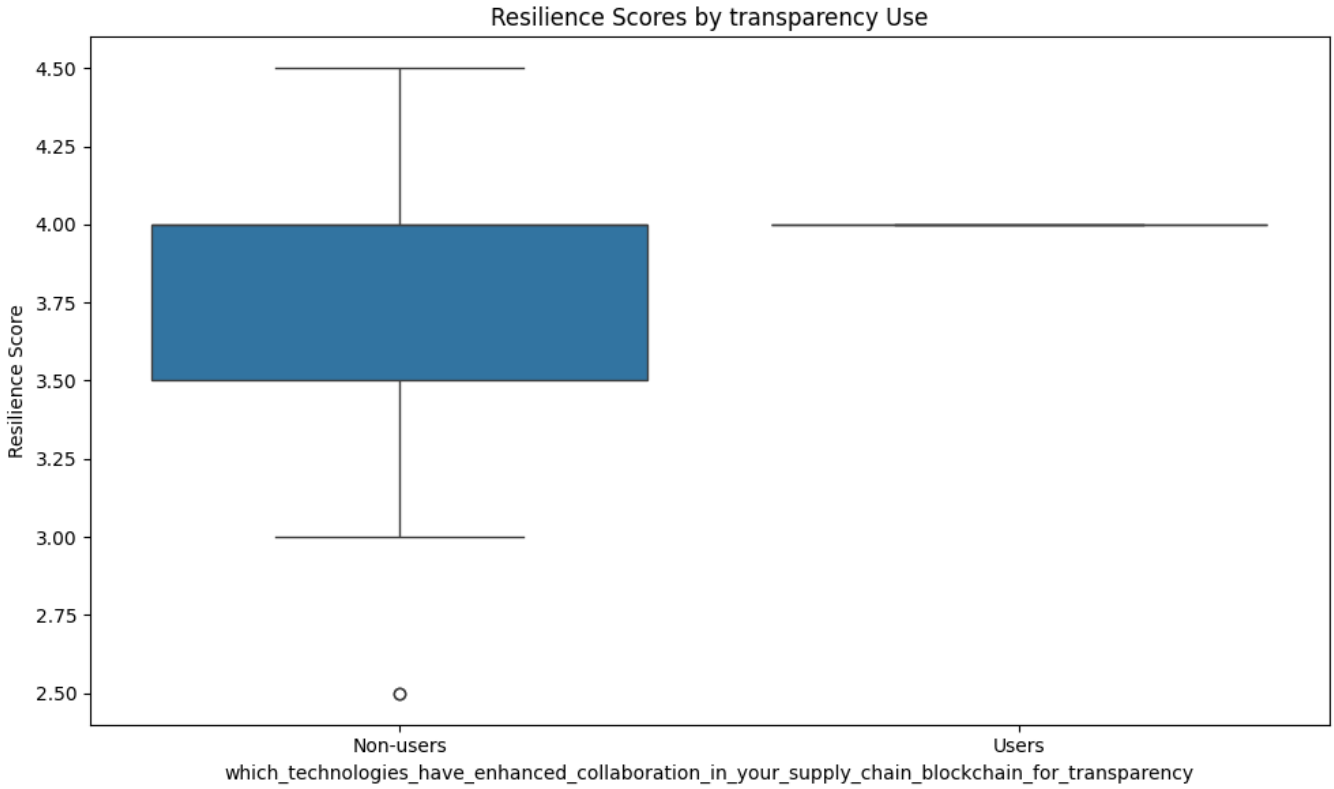


Collaboration Technology: transparency

transparency:

Mann-Whitney U = 14.5, p = 0.623

No significant difference in resilience scores.



Hypothesis H5b

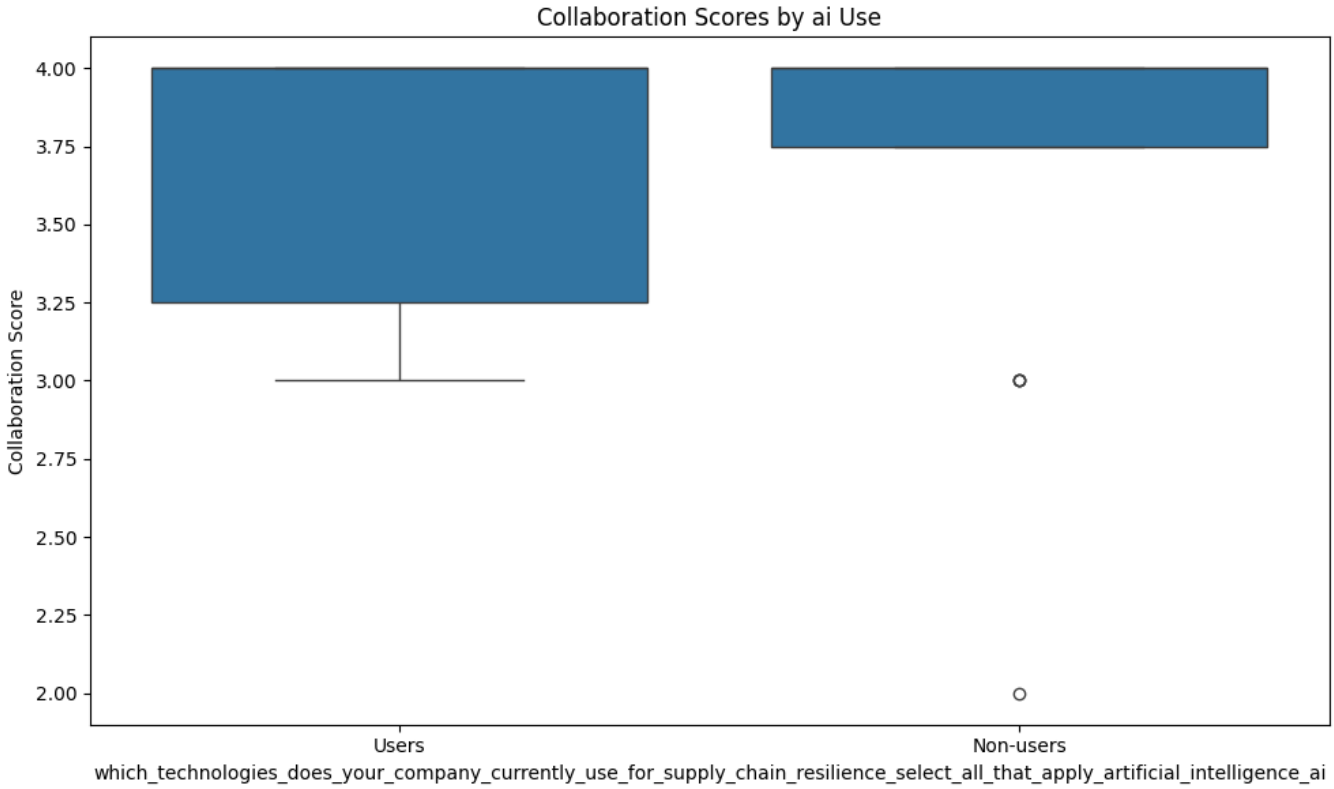
The use of technology strengthens collaboration and trust among stakeholders.

Technology and Collaboration: ai

ai:

Point-Biserial Correlation: r = -0.017, p = 0.941

No significant relationship found.

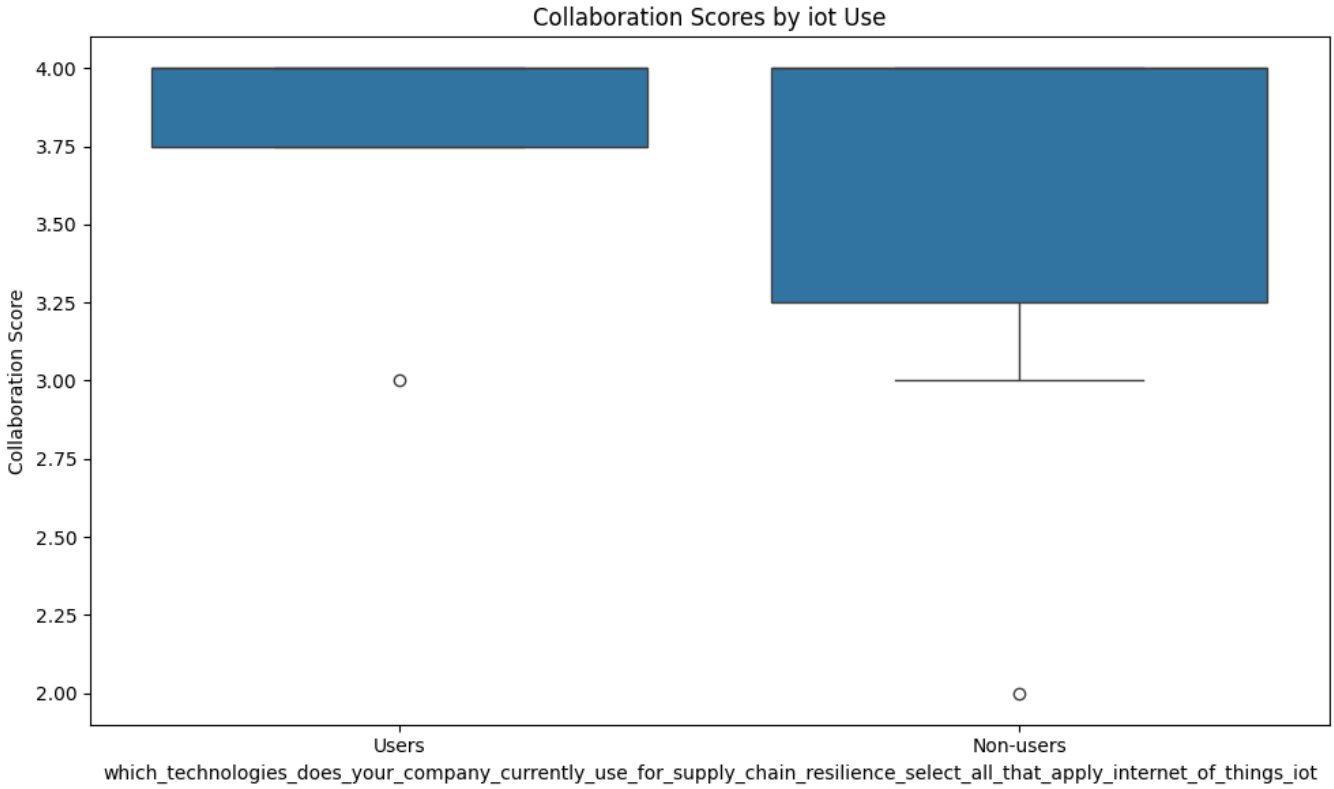


Technology and Collaboration: iot

iot:

Point-Biserial Correlation: r = 0.058, p = 0.798

No significant relationship found.

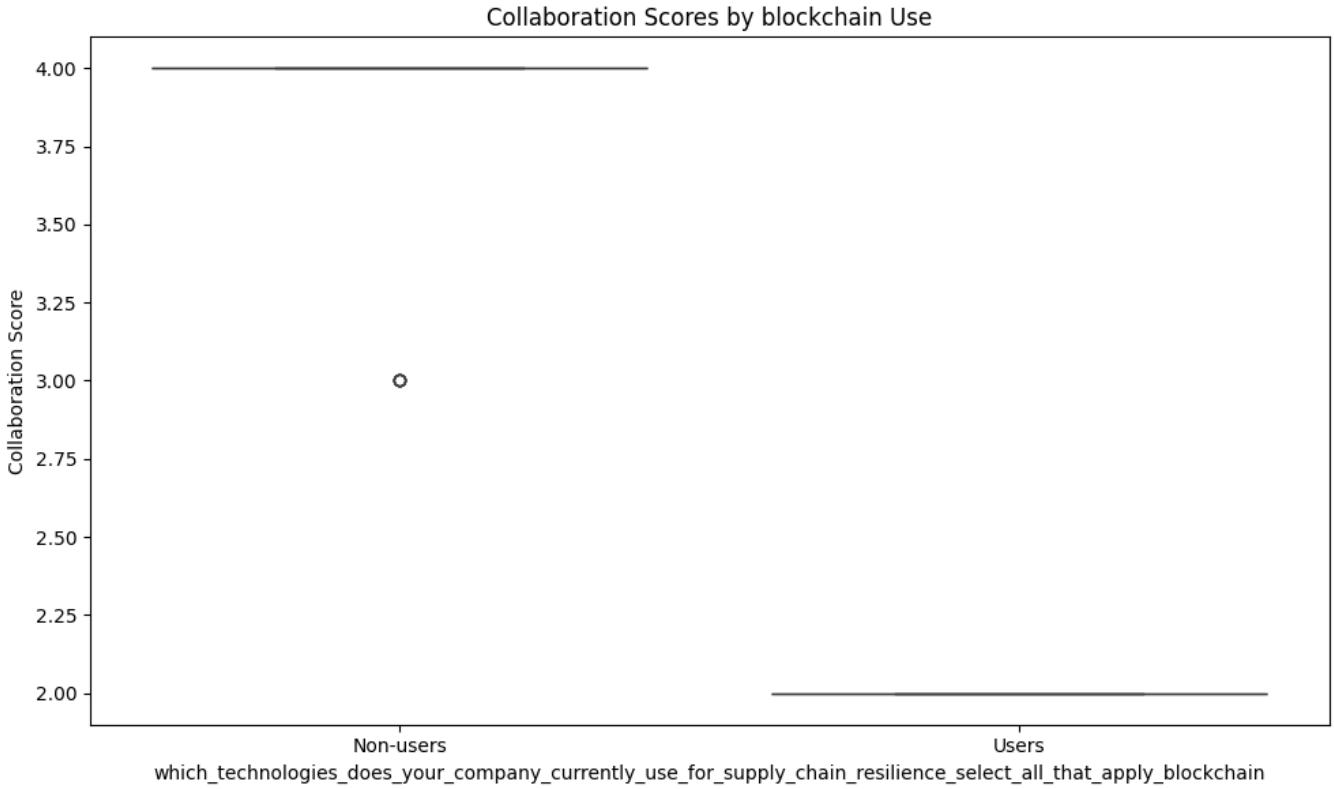


Technology and Collaboration: blockchain

blockchain:

Point-Biserial Correlation: r = -0.661, p = 0.001

Significant positive relationship found.

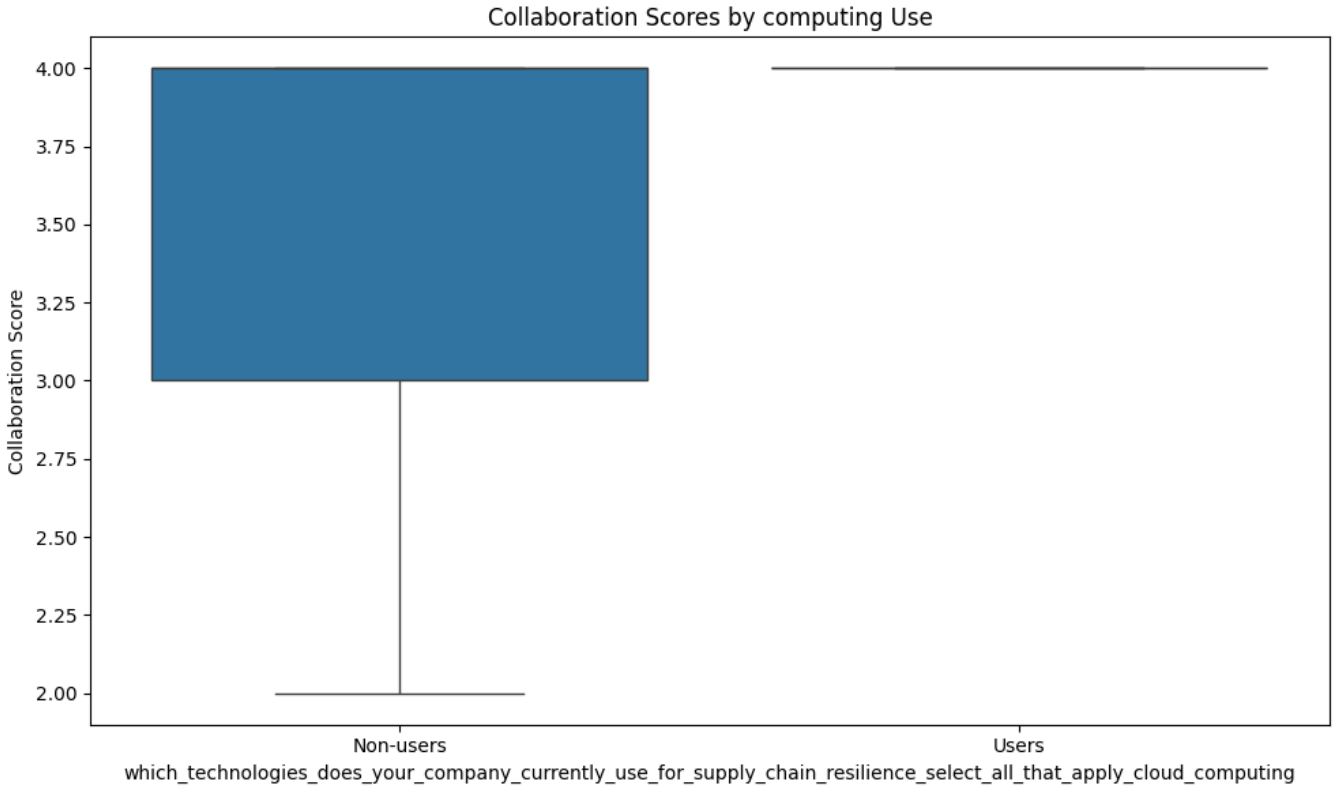


Technology and Collaboration: computing

computing:

Point-Biserial Correlation: r = 0.125, p = 0.579

No significant relationship found.

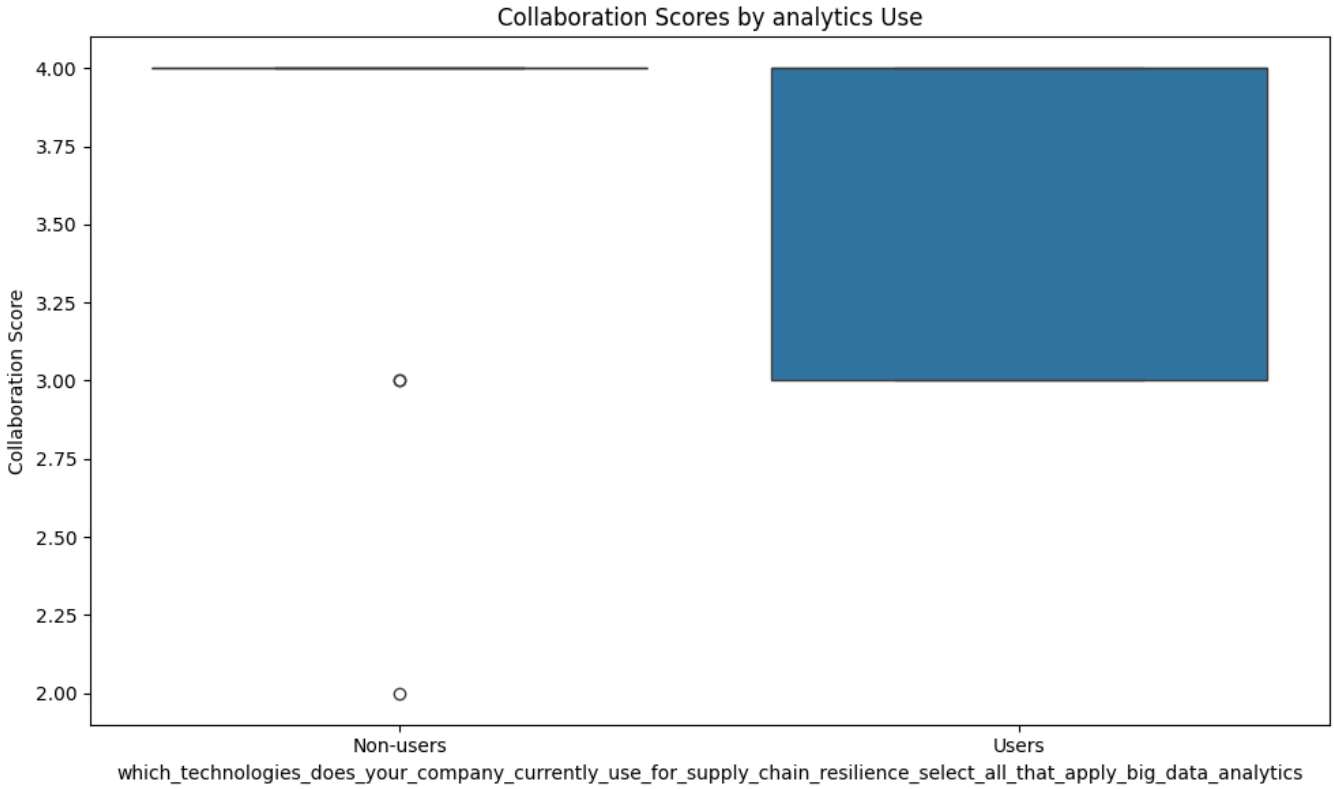


Technology and Collaboration: analytics

analytics:

Point-Biserial Correlation: r = -0.077, p = 0.732

No significant relationship found.

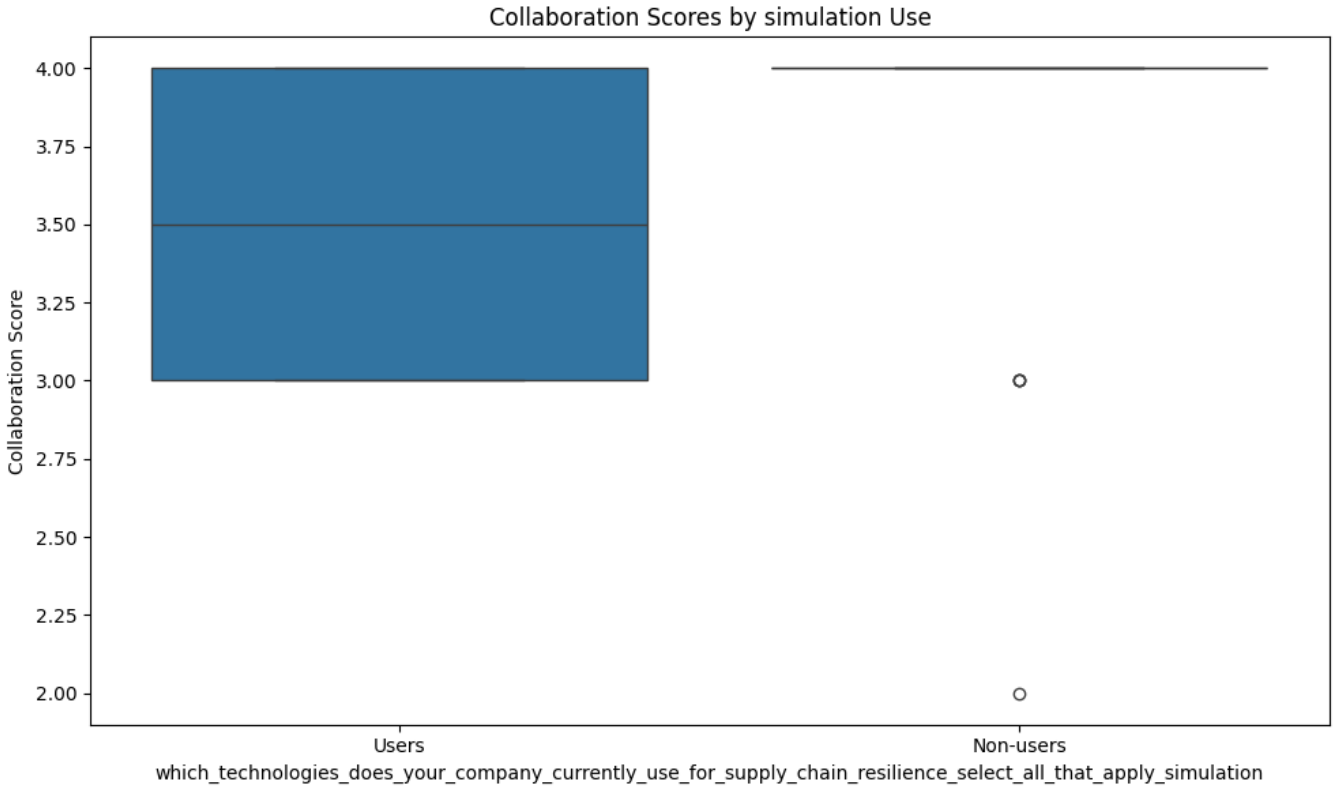


Technology and Collaboration: simulation

simulation:

Point-Biserial Correlation: r = -0.154, p = 0.492

No significant relationship found.



Hypothesis H6a

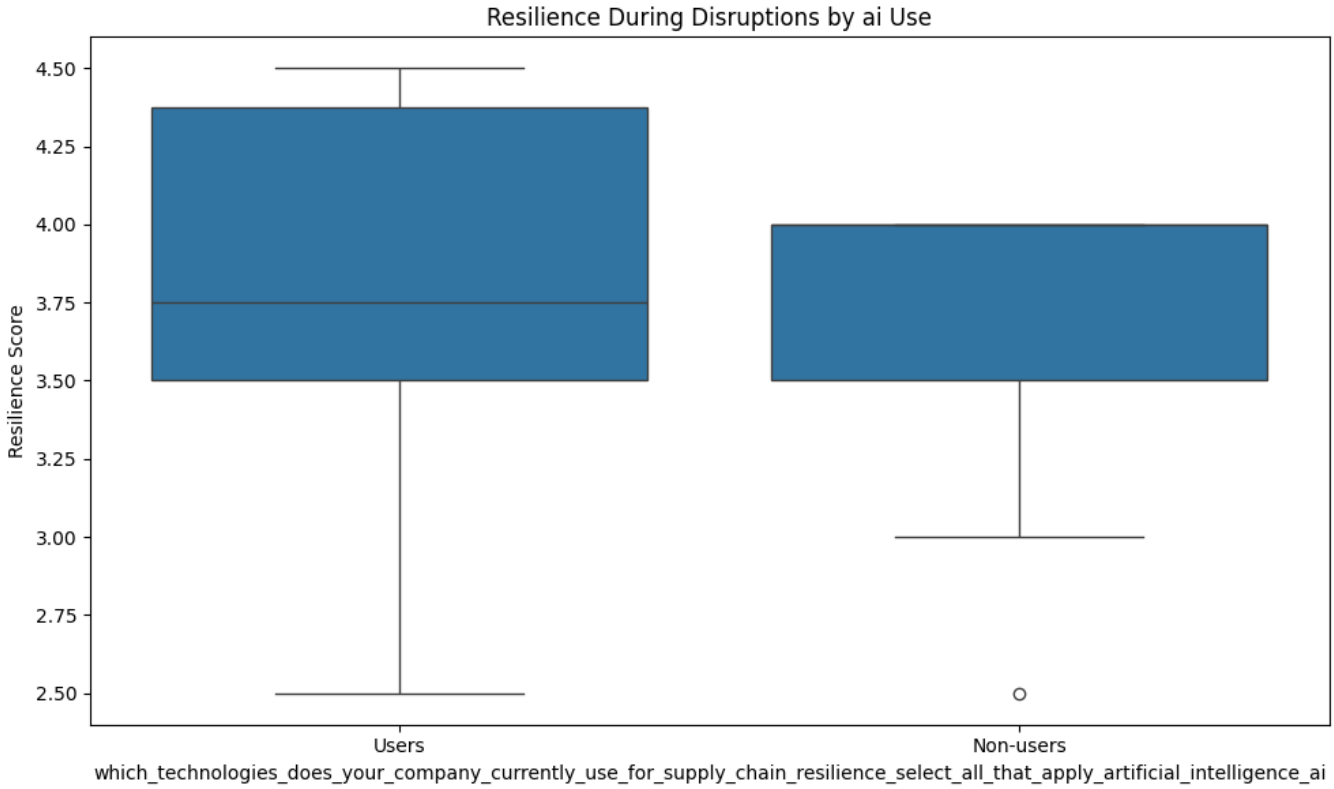
Companies using real-time data analytics and AI-powered decision support systems make more effective supply chain decisions during disruptions.

Decision Technology: ai

ai during disruptions:

Mann-Whitney U = 55.0, p = 0.790

No significant difference in resilience scores during disruptions.

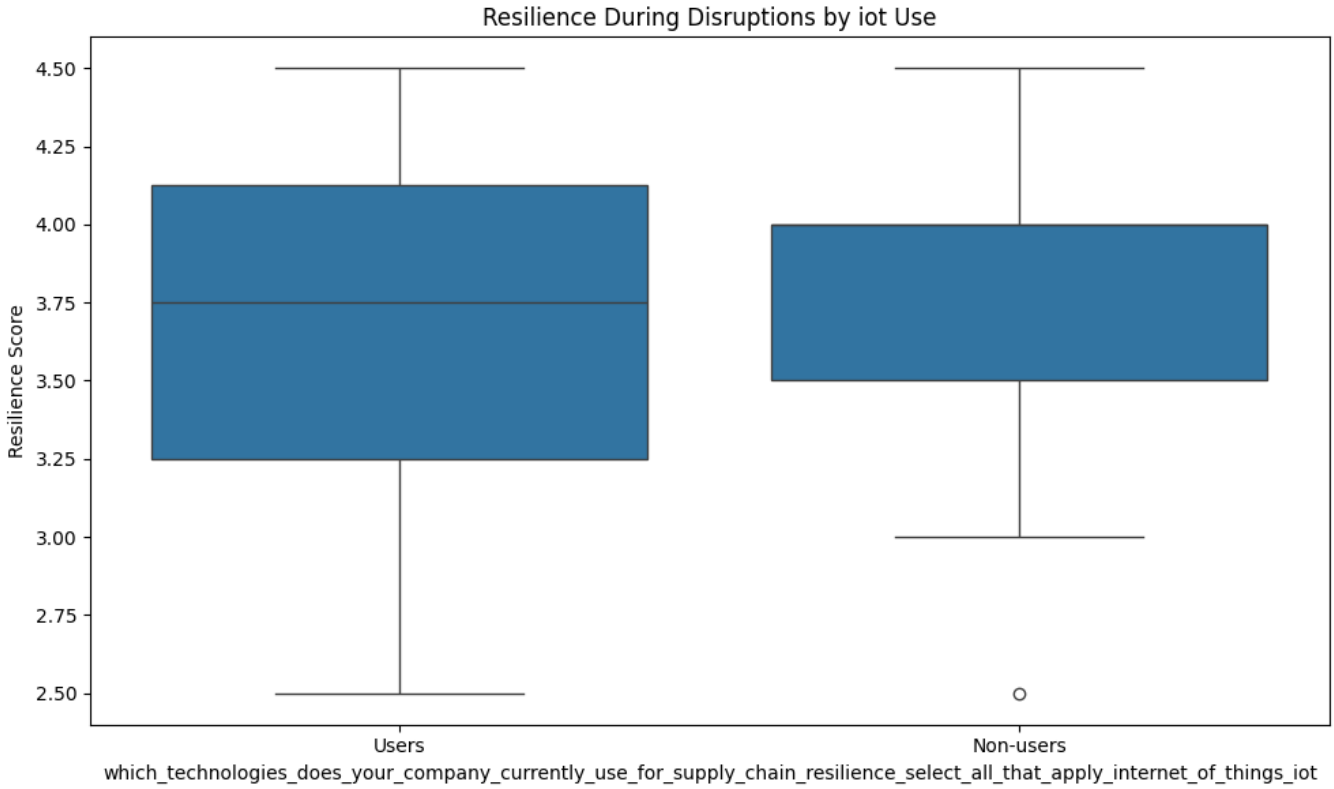


Decision Technology: iot

iot during disruptions:

Mann-Whitney U = 36.5, p = 0.930

No significant difference in resilience scores during disruptions.

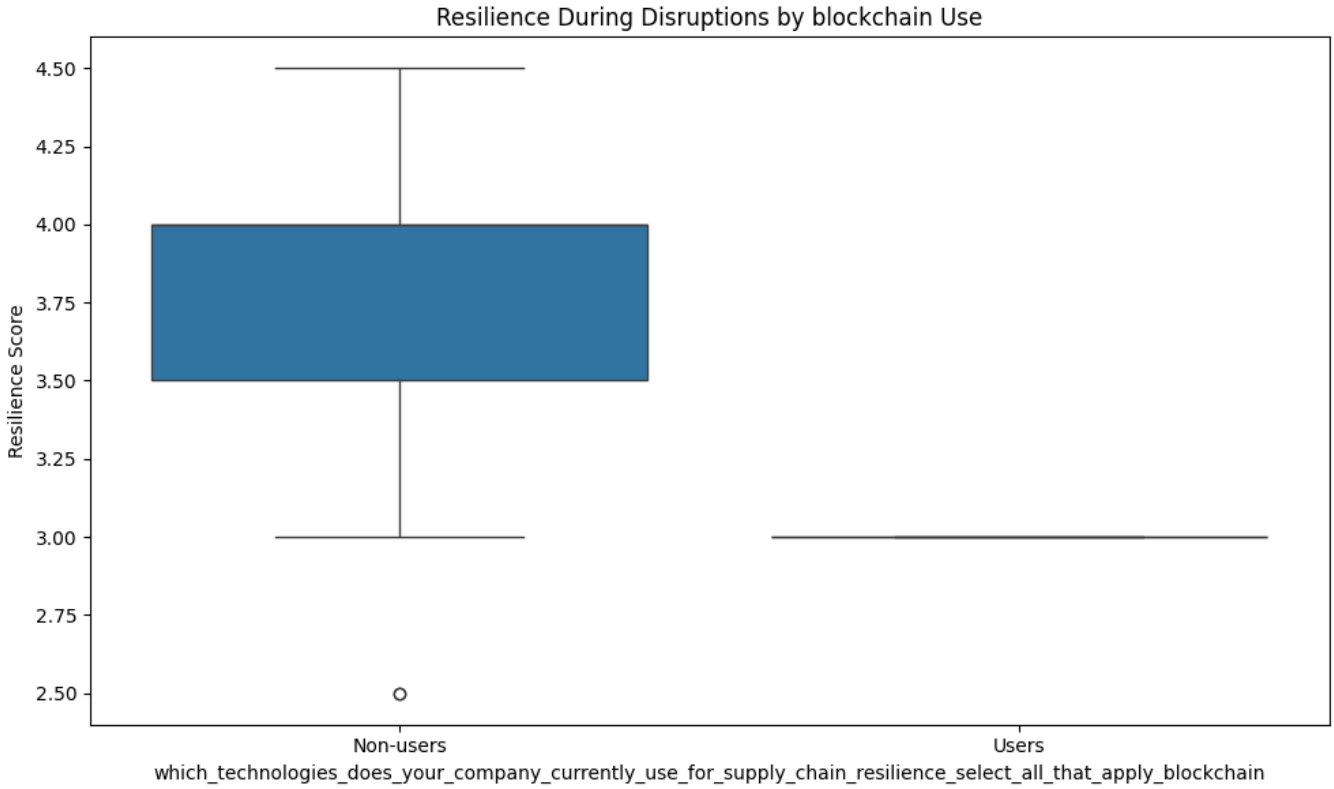


Decision Technology: blockchain

blockchain during disruptions:

Mann-Whitney U = 2.5, p = 0.190

No significant difference in resilience scores during disruptions.

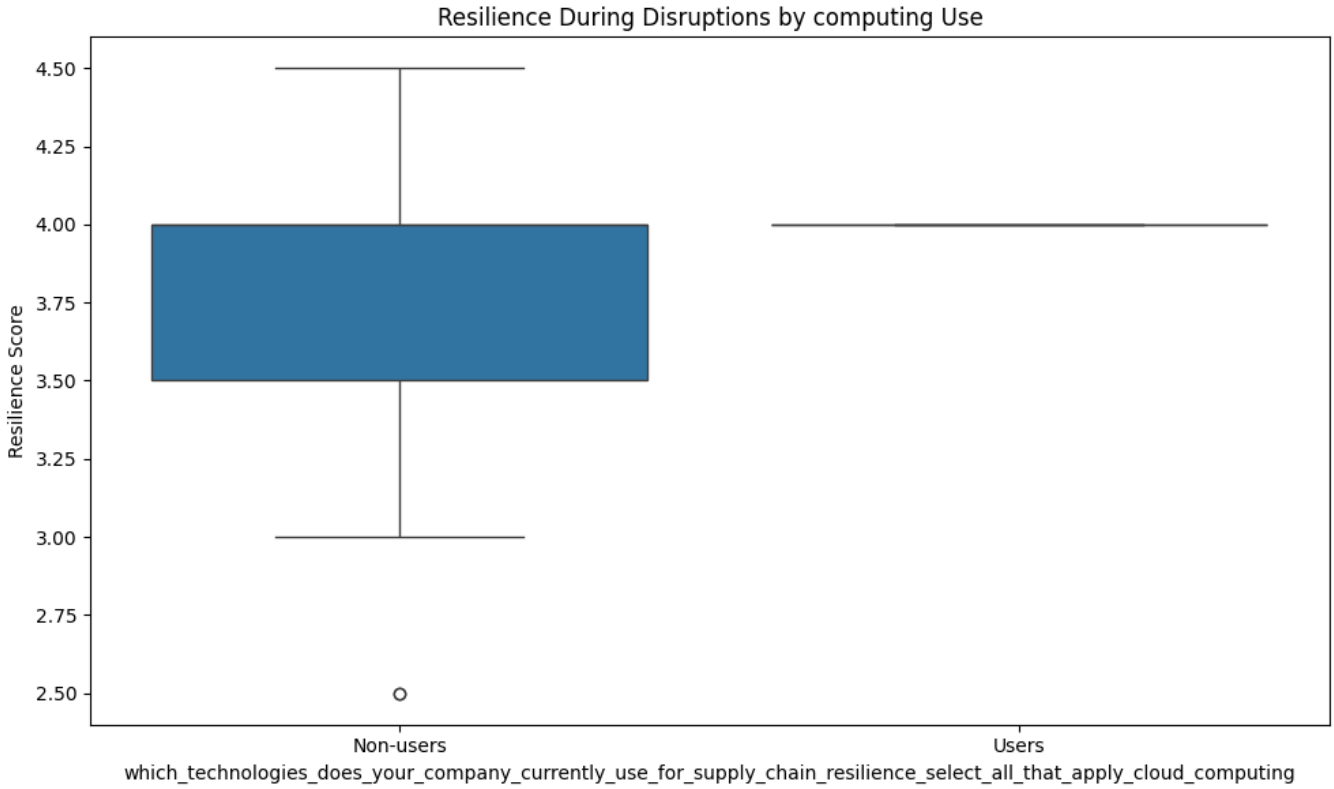


Decision Technology: computing

computing during disruptions:

Mann-Whitney U = 14.5, p = 0.623

No significant difference in resilience scores during disruptions.

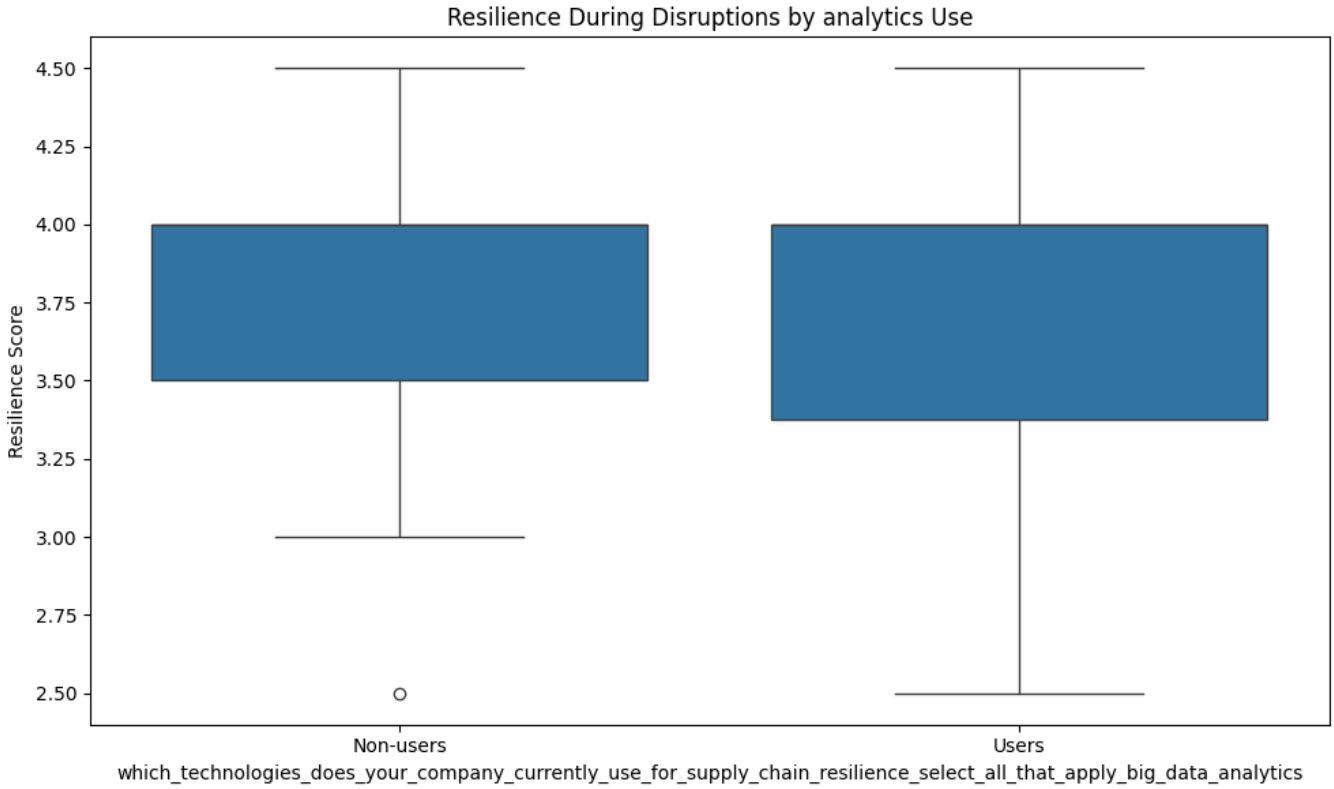


Decision Technology: analytics

analytics during disruptions:

Mann-Whitney U = 60.5, p = 1.000

No significant difference in resilience scores during disruptions.

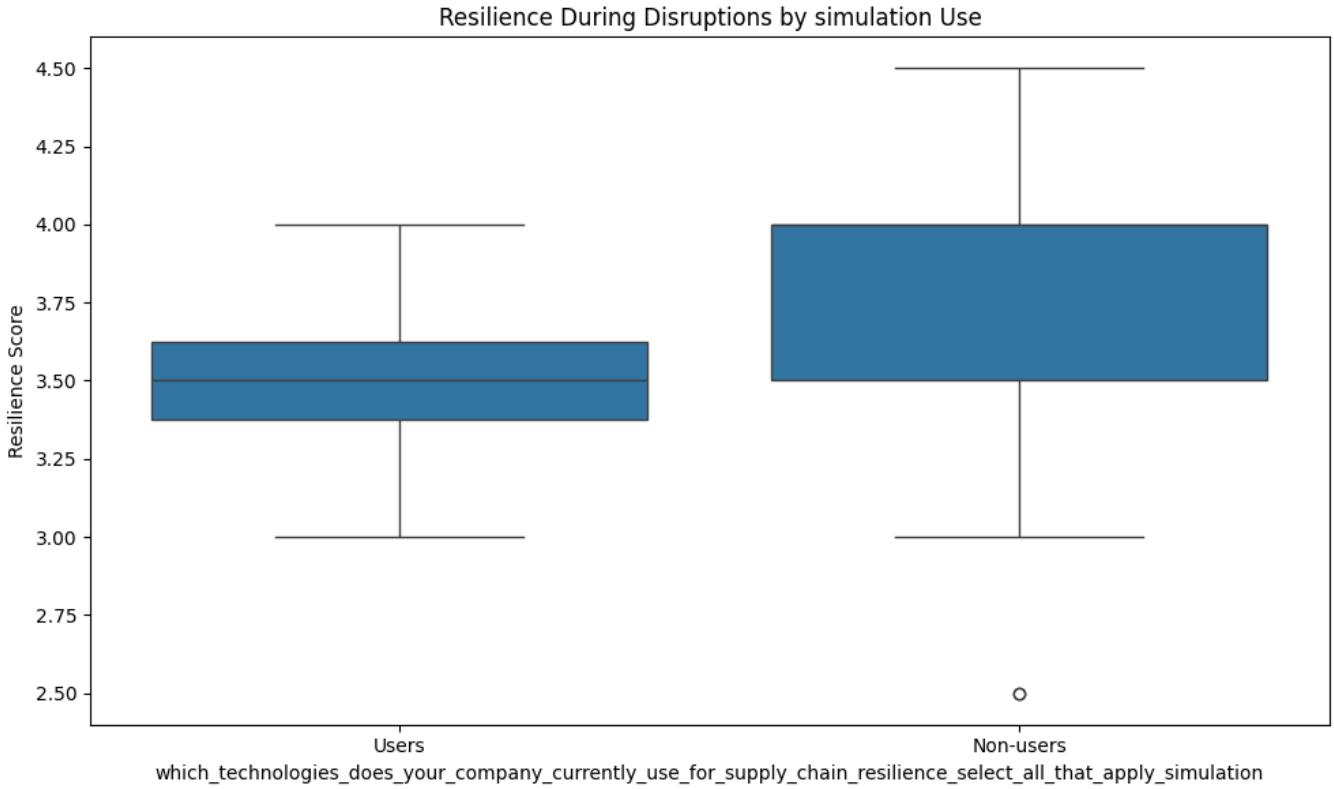


Decision Technology: simulation

simulation during disruptions:

Mann-Whitney U = 23.0, p = 0.201

No significant difference in resilience scores during disruptions.



Hypothesis H6b

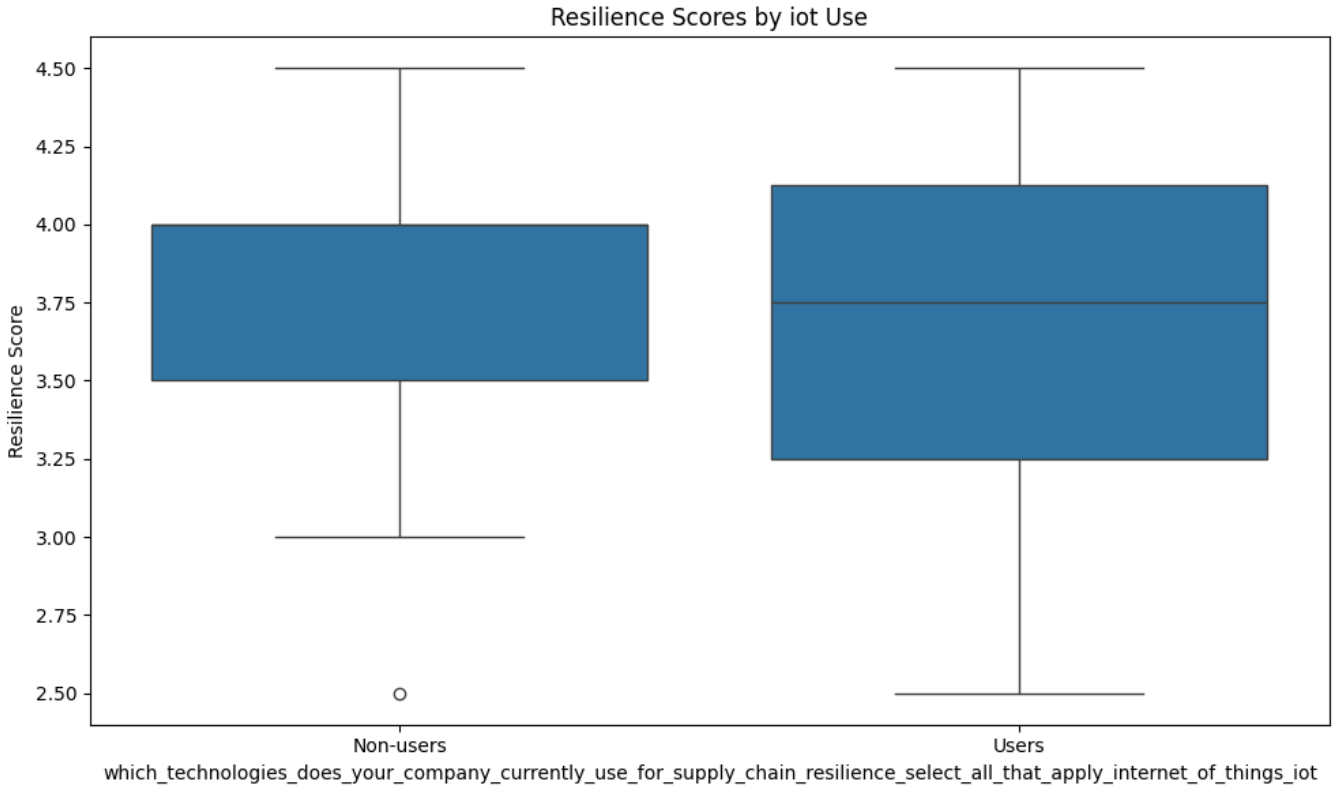
Organizations that implement automated risk assessment tools report improved supply chain visibility and control.

Risk Assessment Technology: iot

iot:

Mann-Whitney U = 36.5, p = 0.930

No significant difference in resilience scores.

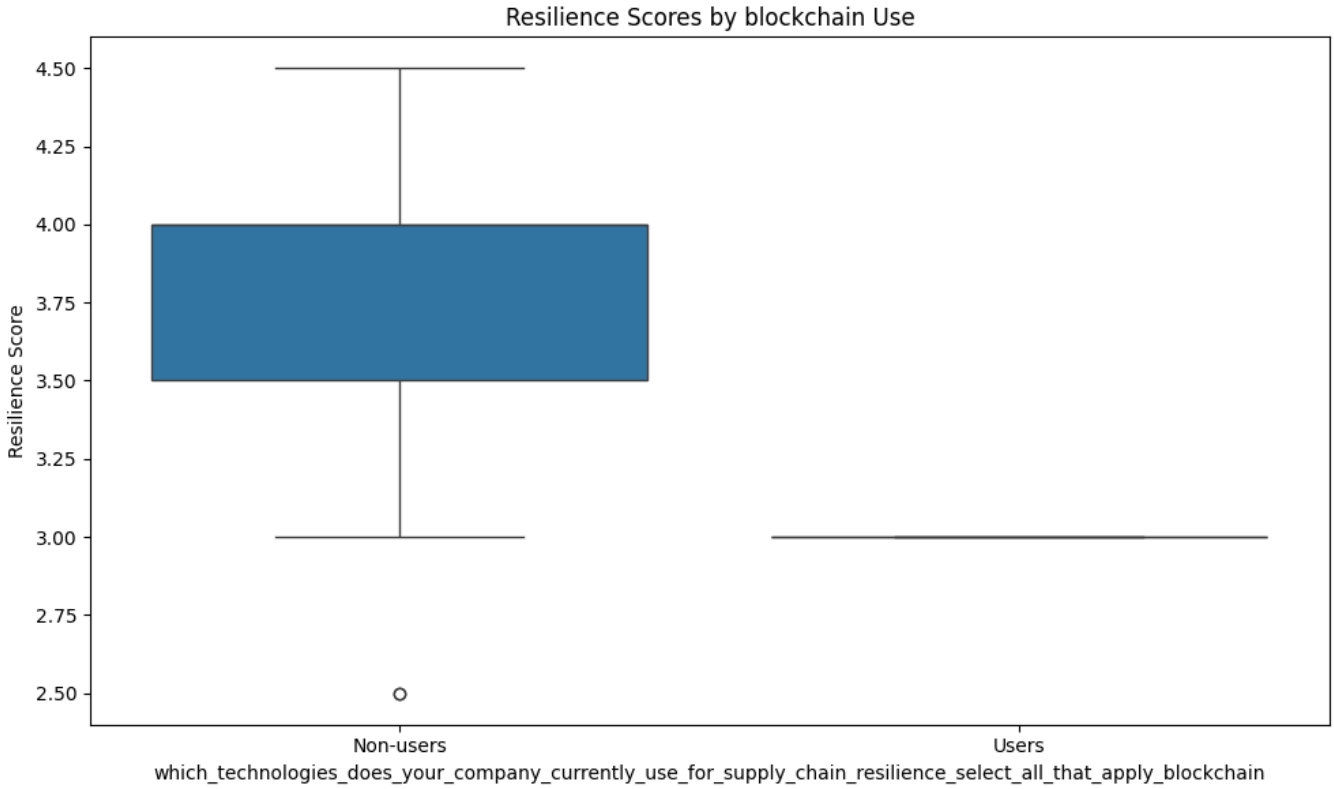


Risk Assessment Technology: blockchain

blockchain:

Mann-Whitney U = 2.5, p = 0.190

No significant difference in resilience scores.



Executive Summary

EXECUTIVE SUMMARY

Key Findings:

1. Disruption and Technology Adoption:

- Companies experiencing more disruptions tend to adopt more technologies ($r=0.38$, $p=0.011$)

- Natural disasters and supplier failures show strongest relationships with specific tech adoption

2. Technology Adoption Patterns:

- Large enterprises adopt AI and Big Data at higher rates than SMEs

- Manufacturing companies show higher adoption of Cloud Computing

3. Technology Effectiveness:

- AI and IoT users report higher resilience scores

- Companies using multiple technologies together show better resilience

4. Barriers to Adoption:

- High implementation costs reported by 45% of companies

- Skill gaps significantly affect AI adoption

Recommendations:

1. For disrupted companies: Prioritize technology adoption, especially AI and IoT

2. For SMEs: Focus on scalable, cost-effective technologies first

3. Address skill gaps through training and partnerships